

SEMESTER 3

AGRICULTURAL ENGINEERING

SEMESTER S3

MATHEMATICS FOR PHYSICAL SCIENCE – 3

(Common to B & C Groups)

Course Code	GYMAT301	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2Hrs. 30 Mins.
Prerequisites (if any)	Basic knowledge in complex numbers.	Course Type	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. (Text 1: Relevant topics from sections 11.7, 11.8, 11.9)	9
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w = z^2$, $w = e^z$, $w = \frac{1}{z}$, $w = \sin z$. (Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)	9

3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, Second evaluation method, Cauchy's integral theorem (without proof) on simply connected domain, Independence of path, Cauchy integral theorem on multiply connected domain (without proof), Cauchy Integral formula (without proof). (Text 1: Relevant topics from sections 14.1, 14.2, 14.3)	9
4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $\cos\theta$ and $\sin\theta$. (Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	K3
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	K3
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	K3
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10 th edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3 rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39 th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44 th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1 st edition, 2011

SEMESTER S3

STRENGTH OF MATERIALS & THEORY OF STRUCTURES

Course Code	PCAGT302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To make the students understand the basic concepts of stress, strain, and elasticity, and their applications in analyzing the behavior of materials under various loading conditions.
2. To develop the ability to analyze the mechanical behavior of structural elements including beams, columns, and shafts subjected to axial, bending, torsional, and combined loading.
3. To provide an insight into the principles of structural design, including selecting appropriate materials, sizing structural members, and ensuring structural safety and serviceability.
4. To make the students learn the methods to analyze deflection and stability of structural elements under different loading conditions, and understand their implications for structural design.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Simple stresses and strains – Concept and types, St. Venant’s principle, stress-strain diagram, Hooke’s law, Elasticity and plasticity, Working stress, Factor of safety, Lateral strain, Poisson’s ratio and volumetric strain, Elastic constants and the relationship between them, Bars of varying section, composite bars. Strain Energy, Resilience – Gradual, sudden, impact and shock loadings and its simple applications. Composite stresses and strains - Two-dimensional system, stress at a point on a plane, principal stresses and principal planes, Mohr’s circle of stress.	11
2	Shear stress and shear strain, Modulus of rigidity, simple shear, punching shear. Beams -different types, Types of loading on beams, Concept of bending moment and shear force, Derivation of relationship between intensity of load, shear force and bending moment, Shear force and bending moment diagrams of cantilever beams, simply supported beams and overhanging beams for different type of loads, Point of contraflexure.	11
3	Bending stresses in beams, theory of simple bending, neutral axis, moment of resistance, bending stresses in beams of various sections, strength of a section. Shear stress in beams -Derivation of the equation for shear stress, Shear stresses in beams of various sections. Torsion, Torsion of circular and hollow circular shafts, Power transmitted by a shaft, strength of a solid shaft and a hollow shaft, derivation of the torsional equations, assumptions, variation of stress across the cross section, Polar modulus. Close and open coiled springs – derivation.	11

4	Deflections of beams – relation between slope, deflection and radius of curvature, Deflection of cantilever and simply supported beams by double integration method, Macaulay's method, moment area method and conjugate beam method. Combined bending and direct stresses, eccentric loading, limit of eccentricity, middle third rule for rectangular and other sections. Condition for no tension in the sections, Stability of masonry dams. Columns and struts, Euler's column theory, derivation for buckling load for four cases of long columns, Rankine's formula for long columns, Slenderness ratio, limitation of Euler's formula, Safe load calculation using Rankine's formula. Statically indeterminate beams, analysis of fixed beams, analysis of continuous beams by Clapeyron's three moment theorem and moment distribution method.	11
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamental terms and theorems associated with strength of materials	K1
CO2	Understand the common modes of structural failure, including yielding, buckling, and fatigue, and be able to assess the factors contributing to failure.	K2
CO3	Analyze the behavior of various structural elements under different loading conditions, including axial, torsional, bending, and combined loading.	K4
CO4	Conduct stress transformations, recognize principal planes and stresses, and determine the maximum shear stress at a specific point within a structural component.	K3
CO5	Design structural elements and systems to meet specified criteria, such as strength, stiffness, stability, and serviceability.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1												3
CO2	3	1	2	1							2	1
CO3	2	2		1								
CO4	2			2								
CO5	3	1	2		2						2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanics of Structures Vol - I	H. J. Shah and S. B. Junnarkar	Charotar Publishing House	24 th Edition, 2015
2	Strength of Materials Vol - I & Vol - II	Stephen Timoshenko	CBS Publishers & Distributors, New Delhi	3 rd Edition, 2010
3	Engineering Mechanics and Strength of Materials	Bansal R. K.	Laxmi Publications, New Delhi.	2014
4	Applied Mechanics and Strength of Materials	Prasad I. B.	Khanna Publishers, New Delhi	2004

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanics of Materials	R. C. Hibbler	Pearson	11 th Edition, 2016
2	Strength of Materials	S. Ramamrutham and R. Narayanan	Dhanpat Rai Publishing Co (P) Ltd.	2017
3	Strength of Materials	S. S. Rattan	McGraw Hill Education India	3 rd Edition, 2017
4	Mechanics of Materials	B. C. Punmia, Ashok K. Jain, Arun Kumar Jain	Laxmi Publications (P) Ltd, New Delhi	2017

Video Links (NPTEL, SWAYAM...)

Module No.	Link ID
1	https://www.youtube.com/watch?v=oDx8vsm9HW8&list=PLJoALJA_KMOARYNi50T6b488kPUBbOIsX&index=18
2	https://www.youtube.com/watch?v=74pm8A0RJ-0
3	https://www.youtube.com/watch?v=uprNzNRcPTg&list=PLJoALJA_KMOARYNi50T6b488kPUBbOIsX&index=27
4	https://www.youtube.com/watch?v=M-0U9E0O8q8&list=PLJoALJA_KMOARYNi50T6b488kPUBbOIsX&index=33

SEMESTER S3

FLUID MECHANICS & OPEN CHANNEL HYDRAULICS

Course Code	PCAGT303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To understand the basic principles and fundamental concepts of fluid flow
2. To impart knowledge on static, kinematics and dynamic aspects of Fluid Mechanics
3. To understand hydraulics of flow through pipes and open channels
4. To acquire knowledge on hydraulic machines such as pumps and turbines

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction and basic concepts; Properties of fluids- density, specific weight, vapour pressure, viscosity, Newton's law of viscosity, surface tension and capillarity, ideal and real fluid; Pressure and its measurement- fluid pressure at a point, Pascal's law, pressure variation in a fluid at a point, gauge pressure, absolute pressure, vacuum pressure; Measurement of pressure-piezometer, manometers -simple U tube, differential and single column manometers and mechanical pressure gauges.; Fluid statics- total pressure and centre of pressure, pressure forces on plane and curved surfaces, buoyancy and floatation- expression for buoyant force and centre of buoyancy, metacentre and meta centric height- determination of metacentric height-experimental and analytical method; Principle of flotation and stability of floating and submerged bodies.	11

2	Kinematics of fluid flow: Eulerian and Lagrangian approaches, types of fluid flow- 1- D, 2-D and 3-D flow, steady, unsteady, uniform, non-uniform, laminar, turbulent, rotational, irrotational flows, translation, rotation, circulation and vorticity, vortex motion-free vortex and forced vortex; continuity equation-continuity equation in three dimensions, velocity and acceleration; streamlines, path lines, streak lines, stream function, velocity potential and flow net; Dynamics of fluid flow – Eulers and Bernoulli's equation of motion, application of Bernoulli's equation-venturi meter, Orifice meter, nozzle meter and pitot tube; Other flow measuring devices-orifices and mouthpieces, notches and weirs-classification, flow through them and hydraulic coefficients; Energies in flowing fluid- pressure head, dynamic, static and total head.	11
3	Flow through pipes – laminar and turbulent flow, head loss due to friction-major energy losses- Darcy Weisbach equation, Chezy's equation, Moodys chart, Minor losses- Head loss due to pipe fittings (only description); Hydraulic gradient and total energy line; flow through network of pipes- pipes in series, pipes in parallel, equivalent pipe; Power transmission through pipes; siphon, Water hammer and Cavitation (Description only). Open channel flow-types of flow in channels, velocity distribution, Measurement of velocity in channels- current meter, parshall flumes, cut-throat flumes and venturi flumes, Chezy's and Manning's equation; Most economical section of channels-rectangular, trapezoidal, triangular and circular channel; Specific energy, critical depth and hydraulic jump.	11
4	Pumps- water lifting devices; different types of pumps, classification of pumps, component parts of centrifugal pumps- priming, pump selection, installation and troubleshooting, performance curves, effect of speed on capacity, head and power, effect of change of impeller dimensions on performance characteristics, work done, efficiencies, cavitations, characteristics curves, specific speed NPSH; Multistage pump, Axila flow / propeller pumps, mixed flow pumps and their performance characteristics; Deep well turbine pump and submersible pump.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand Properties of fluids, classify fluid flows and examine the different pressure measurement methods	K2
CO2	Apply Euler and Bernaulli's equation and investigate the flow measuring devices in pipes and open channels	K3
CO3	Apply the flow through pipes for the design of pipes in various situations.	K3
CO4	Apply the flow through channels for the design of channels in various situations.	K3
CO5	Understand the characteristics of different types of Pumps for their selection in various situations	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2			1			1			
CO2	3	2	2			1			3			
CO3	3	3	3	1		2			2			
CO4	2	1		1					1			
CO5	3	3	3			2			2			

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hydraulics & Fluid Mechanics, including hydraulic machines	Modi P. N. and S. M. Seth	S.B.H Publishers, New Delhi	18 th , 2011
2	A Text Book of Hydraulics, Fluid Mechanics and Hydraulic Machines	Khurmi,R .S.	S. Chand & Company Limited, New Delhi.	15 th , 1987
3	Fluid Mechanics: fundamentals and Applications	Cengel and Cimbala	McGraw Hill Education India	4 th , 2019
4	Introduction to Fluid Mechanics	Fox and McDonald	John Wiley and Sons	8 th , 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Textbook of Fluid Mechanics and Hydraulic Machines,	Bansal R. K.,	Firewall Media	10 th , 2019
2	Fluid Mechanics: An Introduction	Radhakrishnan, E	Prentice Hall India,	4 th , 2019
3	Mechanics of Fluids	Shames I. H	McGraw Hill	3 rd , 2013
4	Fluid Mechanics	White F.M	Tata McGraw Hill	9 th , 2022

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/OwC5Dmtau98
2	https://youtu.be/c5lXIESO0zY
3	https://youtu.be/kaF7S5HU-C4 , https://youtu.be/CAZNluQTE-0
4	https://youtu.be/JQmkQQVYnJU

SEMESTER S3

SURFACE AND GROUNDWATER HYDROLOGY

Course Code	PBAGT304	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the students with the important aspects of watershed hydrology.
2. To impart knowledge about the various hydrologic phenomena and their relevance in the field of soil and water conservation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Hydrologic cycle, precipitation and its forms, rainfall measurement and estimation of mean rainfall, frequency analysis of point rainfall. Mass curve, hyetograph, depth-area-duration curves and intensity-duration-frequency relationship. Hydrologic processes-interception, infiltration –factors influencing, measurement and indices.	9
2	Evaporation - estimation and measurement, Runoff - factors affecting, measurement, stage-discharge rating curve, estimation of peak runoff rate and volume, Rational method, Cook's method and SCS curve number method. Hydrograph - components, base flow separation, unit hydrograph theory, S-curve, synthetic hydrograph, applications and limitations.	9

3	Floods – terms and definitions head water flood control- method. Stream gauging - discharge rating curves, flood peak, design flood and computation of probable flood, Flood routing – channel and reservoir routing. Drought – classification causes and impacts, drought management strategy.	9
4	Occurrence and movement of ground water; aquifer and its types; Specific yield and storage coefficient. Steady unidirectional flow in an unconfined and confined aquifer. Steady radial flow to a well in an unconfined and confined aquifer. Surface investigation of groundwater: Geologic methods, Geophysical exploration, electrical resistivity method. Subsurface investigation of groundwater: Test drilling, Water level measurement, Geophysical logging, electrical resistivity logging.	9

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the key drivers on water resources, hydrological processes and their integrated behaviour in catchments	K2
CO2	Explain the runoff characteristics of a watershed and estimate the streamflow	K4
CO3	Understand the different components of hydrograph and its types, applications and limitations	K2
CO4	Study the principles of ground water hydrology and apply them for computing yield of aquifers	K3
CO5	Explain the natural disasters like floods and droughts that affect agricultural land	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2	1								2
CO2	3	2		1	1		1					
CO3	3	2	1	1					1			2
CO4	3	2	1	1	1		1					
CO5	3		1				1					2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Hydrology	Subramanya, K	Tata McGraw-Hill Publishing Co., New Delhi.	5 th Edition, 2020
2	Watershed Hydrology	Suresh, R	Standard Publishers Distributors, New Delhi-6	2 nd Edition, 2005
3	Hydrology: Principles Analysis and Design.	Raghunath, H.M	New Age International (P) Limited Publishers, New Delhi.	2 nd Edition, 2006
4	Hydrology and Soil Conservation Engineering, Including Watershed Management	Ghanshyam Das	Prentice Hall India Learning Private Limited	2 nd Edition, 2009
5	Water well and pump engineering	A.M.Michael and S.D.Khepar	Tata McGraw-Hill publishing Company Ltd. New Delhi	9 th Edition, 2005
6	Groundwater hydrology	David Keith Todd and Larry W. Mays	John Wiley and sons, New York (International Book Disturbing Co. Lucknow)	3 rd Edition, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Applied Hydrology	Chow, V.T., D.R. Maidment and L.W. Mays.	McGraw Hill Publishing Co., New York	2 nd Edition, 2010
2	Land and Water Management Engineering,	V.V.N. Murthy	Kalyani Publishers, New Delhi	6 th Edition, 2013
3	Irrigation Engineering and Hydraulic Structures	S.K.Garg	Khanna Publisher	38 th Edition, 2023

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	GNEST305	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to AI and Machine Learning: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	11

2	Mathematical Foundations of AI and Data science: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	11
3	Applied Probability and Statistics for AI and Data Science: Basics of probability-random variables and statistical measures - rules in probability-Bayes theorem and its applications- statistical estimation-Maximum Likelihood Estimator (MLE) - statistical summaries- Correlation analysis-linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	11
4	Basics of Data Science: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	11

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	K3
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	K3
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	K3
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								
CO2	3	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3								
CO5	3	3	3	3								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6 th edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media, Inc.	2 nd edition, 2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1 st edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9 th edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1 st edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2 nd edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1 st edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1 st edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1 st edition, 2019
5	Probability and Statistics -The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1 st edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmnnnibpcajpcglefindmkaj/ https://www.math.arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender	6

	stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and	6

	human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

SURVEYING AND LEVELLING

Course Code	PCAGL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Practical

Course objectives:

1. To introduce various land surveying equipments and methods
2. To impart concept and skills of various land surveying techniques
3. To impart concept and skills of levelling
4. To plot the result of surveying field works to form maps
5. To impart concept and skills of total station surveying

Details of Experiment

Expt. No	Experiment
1	Chain Survey I– Study of instruments used for chain survey chain, Ranging, Measurement of distance and area
2	Chain Survey II- Triangulation and Traversing
3	Cross – staff survey
4	Compass survey
5	Plane Table Survey I- Study of instruments, Radiation, Intersection, traversing

6	Plane Table Survey II- 2 Point problem and 3 Point problem
7	Levelling I– Study of instruments, Height of collimation method, Rise and fall method
8	Levelling II- Profile and Cross section
9	Levelling III- Contouring
10	Theodolite survey I – Study of the instrument, Measurement of horizontal angle – Repetition method
11	Theodolite survey II- Measurement of horizontal angle – Reiteration method
12	Theodolite survey III- Measurement of vertical angle
13	Tacheometric survey I- Study of instruments, Horizontal distance measurement
14	Tacheometric survey II- Measurement of height of the object, - Measurement of elevation difference between points and the gradient between them
15	Total station survey I
16	Total station survey II

Note: A minimum of 10 experiments are mandatory.

Course Assessment Method (CIE: 50 Marks, ESE 50 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record. (Continuous Assessment)	Internal Exam	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Mandatory requirements for ESE:

- Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- Endorsement by External Examiner: The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course the student will be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Use conventional surveying tools such as chain and compass for field surveying	K1
CO2	Apply levelling principles in field	K3
CO3	Understand the procedure for using plane table surveying	K2
CO4	Employ theodolite for field surveying	K3
CO5	Employ total station for field surveying	K3

K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3							1	2			1
CO2	3			1				1	2			1
CO3	3			1				1	2			1
CO4	3			1	3			1	2			2
CO5	3				3			1				2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Surveying and Levelling.	Kanetkar T P.	Pune Vidyarthi Griha, Prakashan, Pune.	1993.
2	Surveying (Vol.I).	Punmia, B C	Laxmi Publications, New Delhi.	1987
3	Surveying and levelling	Rangwala	Charotar Publishing House Pvt. Ltd. Gujarat.	1991.
4	Surveying(Vol.I)	Arora K R	Standard Book House, Delhi.	1990

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

STRENGTH OF MATERIALS

Course Code	PCAGL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To familiarize students with various material properties such as elasticity, yield strength, modulus of elasticity, etc.
2. To enable students to analyze and evaluate the behavior of structural components under different loading conditions.
3. To bridge the gap between theoretical knowledge gained in lectures and its practical application in real-world scenarios.

Expt. No.	Experiments
1	Determination of hardness of the given specimen by Brinell's Hardness test.
2	Determination of hardness of the given specimen by Rockwell Hardness test.
3	Determination of impact load on a specimen by Izod impact Test.
4	Determination of impact load on a specimen by Charpy's impact test.
5	Determination of modulus of elasticity, ultimate stress, permissible stress, percentage elongation and fracture properties of M. S. and TMT specimens by conducting Tension test in Universal Testing Machine.
6	Determination of compressive strength of concrete cylinders and concrete cubes using compression testing machine.

7	Determination of the various physical and mechanical properties of M.S. Girder, Timber beam, R.C.C. beam etc by performing the bending test in Universal Testing Machine.
8	Determination of ultimate shear strength of mild steel by double shear test.
9	Determine modulus of elasticity of beam using Maxwell's reciprocal theorem.
10	Determination of Modulus of Rigidity of mild steel specimen by performing the torsion test.
11	Analysis of the load deflection characteristics of close coiled spring (tension spring) using spring testing machine.
12	Analysis of the load deflection characteristics of open coiled spring (compression spring) using spring testing machine.
13	Determination of workability of concrete by slump test.
14	Determination of workability of concrete by compaction factor test.
15	Determination of initial and final setting time for cement.
16	Determination of the fatigue behaviour of a specimen by using Fatigue test machine.

*Minimum of 10 experiment

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Gain familiarity with the operation and calibration of laboratory equipment used for material testing, such as universal testing machines, extensometers, and hardness testers.	K2
CO2	Develop the ability to analyze experimental data obtained from material testing, including stress-strain curves, load-displacement curves, and failure modes, and draw meaningful conclusions from the results.	K4
CO3	Acquire hands-on experience in determining mechanical properties of materials through experimentation, including tensile strength, yield strength, modulus of elasticity, workability and hardness.	K3
CO4	Understand the relationship between experimental observations and theoretical concepts learned in the classroom, and recognize the limitations and assumptions of theoretical models compared to real-world behavior.	K2
CO5	Demonstrate a commitment to laboratory safety protocols and practices, including proper handling of equipment, use of personal protective equipment (PPE), and adherence to standard operating procedures.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2		2	2								2
CO2	3	2		2								2
CO3	2	1	2	2								2
CO4			1								1	1
CO5												1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Strength of Materials (Mechanics of Solids)	Rajput R.K.	S.Chand & company Ltd., New Delhi	2018
2	Strength of Materials	Rattan.S.S.	Tata McGraw Hill Education Pvt. Ltd., New Delhi	2017
3	Theory of Structures (SMTS) Vol -II	Punmia B.C., Ashok Kumar Jain and Arun Kumar Jain	Laxmi Publishing Pvt Ltd, New Delhi	2017

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Strength of Materials (Mechanics of Solids)	Rajput R.K.	S.Chand & company Ltd., New Delhi	2018
2	Strength of Materials	Rattan.S.S.	Tata McGraw Hill Education Pvt. Ltd., New Delhi	2017
3	Theory of Structures (SMTS) Vol -II	Punmia B.C., Ashok Kumar Jain and Arun Kumar Jain	Laxmi Publishing Pvt Ltd, New Delhi	2017

Video Links (NPTEL, SWAYAM...)	
Sl. No.	Link ID
1	https://sm-nitk.vlabs.ac.in/exp/izod-impact-test/
2	https://sm-nitk.vlabs.ac.in/exp/bending-test-mild-steel/
3	https://sm-nitk.vlabs.ac.in/exp/tensile-test-mild-steel/
4	https://sm-nitk.vlabs.ac.in/exp/torsion-test-mild-steel/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

AGRICULTURE ENGINEERING

SEMESTER S4

MATHEMATICS FOR PHYSICAL SCIENCE – 4

(Group C)

Course Code	GCMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Mins.
Prerequisites (if any)	Basic calculus.	Course Type	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To provide the students with the basics of various numerical methods to develop problem solving skills used in various engineering disciplines.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. [Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]	9
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables. [Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]	9

3	Confidence Intervals, Confidence Level, Confidence Intervals and One-side confidence intervals for a Population Mean for large and small samples (normal distribution and t -distribution), Hypotheses and Test Procedures, Type I and Type II error, z Tests for Hypotheses about a Population Mean (for large sample), t Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. [Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]	9
4	Newton-Raphson Method, Gauss Elimination Method, Gauss - Jordan Method, Numerical solution of ordinary differential equations-Euler's method, Modified Euler's method, Runge - Kutta method of 2 nd Order, Numerical solution of Laplace equation –Jacobi's Method, Curve Fitting by Method of Least Squares - Straight lines, Parabola. (Text 2: Relevant topics from sections 2.5, 4.2, 7.5, 8.4, 8.5, 9.4)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	K3
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	K3
CO3	Estimate population parameters, assess their certainty with confidence intervals, and test hypotheses about population means and proportions using z -tests and the one-sample t -test.	K3
CO4	Apply numerical methods to find solutions of linear system of equations, ordinary differential equations and Laplace equations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9 th edition, 2016
2	Introductory Methods of Numerical Analysis	S S Sastry	PHI Learning Pvt Limited	5 th edition, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.,	McGraw Hill.	4 th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6 th edition, 2020
3	Numerical methods for Engineers	Steven C. Chapra, Raymond P. Canale	McGraw Hill Education	8 th edition, 2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
3	https://archive.nptel.ac.in/courses/117/105/117105085/
4	https://archive.nptel.ac.in/courses/111/107/111107105/

SEMESTER S4

SOIL MECHANICS AND FOUNDATIONS

Course Code	PCAGT402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the importance of soil mechanics in engineering, index properties and its determination.
2. To understand effective pressure of soil, permeability and compressibility of soil
3. To analyse stress distribution, shear strength and lateral earth pressure of soil
4. To evaluate stability of infinite and finite slope
5. To understand the bearing capacity of soil and different kinds of foundations used in engg constructions

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Soil engineering and index properties: Soil and soil engineering, history of soil mechanics, Soil as a 3 phase system, Index properties: void ratio - specific gravity - dry density - porosity - water content - saturated unit weight - submerged unit weight - degree of saturation. Determination of index properties of soils: Determination of water content by oven drying–Specific gravity using pycnometer and specific gravity bottle - Grain size analysis by sieve analysis, hydrometer analysis and pipette analysis - Atterberg limits and	11

	indices – Visual identification by simple field tests – Field density by core cutter and sand replacement methods. Classification of soils: Necessity - Principles of classification - I.S. classification – Plasticity charts – Group index.	
2	Soil water: Modes of occurrence – adsorbed and capillary water types - Total stress -Effective stress – Pore pressure - Pressure diagrams. Permeability: Definition - Darcy’s law - Factors affecting permeability – Laboratory determination - Stratified soils: average permeability. Compaction: Definition and objectives of compaction – Standard and Modified Proctor tests- Concept of OMC and maximum dry density - Zero air voids line - Factors influencing compaction - Effect of compaction on soil properties - Field compaction methods – Proctor needle for field control. Consolidation: Definition –Spring analogy for primary consolidation - Terzaghi’s theory of one dimensional consolidation – Concepts of coefficient of compressibility - Coefficient of volume change and compression index – Laboratory consolidation test - e-log p curves.	11
3	Stress distribution in soil: Stress due to self weight, concentrated load - Boussinesq equations, pressure distribution diagrams, vertical pressure under a uniformly loaded circular area. Shear strength: Definition - Mohr’s strength and stress circles - origin of planes - Mohr’s envelope - Mohr- Coulomb strength theory –Direct shear test – introduction to triaxial shear test- Measurement of pore pressure -Total and effective stress strength parameters - UCC test - Vane shear tests - Choice of test conditions for field problems. Lateral earth pressure: Earth pressure at rest - Active and passive earth pressure for cohesionless and cohesive soils - Rankine’s and Coulomb’s theories - Point of application of earth pressure for cases of with and without surcharge in cohesionless and cohesive soils - Culmann’s and Rebhan’s graphical construction	11

	for active earth pressure- Stability of slopes: Slope failure, base failure and toe failure - Swedish circle method – $\Phi = 0$ analysis and $c = 0$ analysis - Friction circle method - Taylor's stability number -Stability charts.	
4	Bearing capacity of soils: Definitions related to foundations and bearing capacity, Minimum depth of foundation - Rankine's analysis, type of bearing capacity failures, effect of water table on bearing capacity, IS code method for computing bearing capacity. Shallow foundations: type of foundations, Spread footing, Safe bearing pressure, Settlement of footings. Pile foundations: type of piles, Pile driving, load carrying capacity of piles, dynamic formulae, Static formulae, Pile load tests, Penetration tests. Well foundations: Caissons, Shape of wells and component parts, Depth of well foundation and bearing capacity, forces acting on a well foundation.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the importance of soil mechanics in engineering, index properties and its determination.	K2
CO2	Understand the concept of effective pressure of soil, permeability and compressibility of soil	K2
CO3	Analyse stress distribution, shear strength and lateral earth pressure of soil	K3
CO4	Evaluate stability of infinite and finite slope	K3
CO5	Understand the bearing capacity of soil and different kinds of foundations used in engineering constructions	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	2							1	
CO2	3	3	1	1								
CO3	3	3	1									
CO4	2	3	2	2								
CO5	2	3	2	2							1	

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Text Book of Soil Science.	Biswas, T.D. and Mukherjee, S.K.	Tata McGraw Hill Publishing Co., New Delhi	
2	. Introductory Soil Science. .	Das.D.K	Kalyani Publishers, New Delhi	
3	Nature and Properties of Soils.	Brady, N.C	Pearson	

SEMESTER S4

THEORY OF MACHINES

Course Code	PCAGT403	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	AGT306	Course Type	Theory

Course Objectives:

1. To identify and enumerate different mechanisms and analyse velocity and accelerations of the links involved in the mechanism.
2. To familiarize design and strength requirements of various power transmission elements including cam and follower, belt drives, bearings, coupling, keys, joints and bolts.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Links and Mechanisms: Definitions Link or Element, Kinematic Pairs, Degrees of Freedom, Grubler's Criterion (without derivation), Kinematic Chain, Mechanism, Structure, velocity and acceleration diagram of links in mechanism (acceleration diagram description only), Inversion, Machine. Static force analysis: Introduction: Static equilibrium. Equilibrium of two and three force members. Members with two forces and torque, free body diagrams, principle of virtual work. Force principle: D'Alembert's Principle, Inertia force, inertia torque, dynamic equivalent system, Dynamic force analysis of four-bar mechanism and slider crank mechanism (explanation only).	11

2	Design of flat belt- material for belts, slip of the belts, creep, centrifugal tension, Power transmitted by flat belts, Design procedure of flat belts. Balancing of Rotating Masses: Static and dynamic balancing, balancing of single rotating mass by balancing masses in same plane and in different planes. Balancing of several rotating masses by balancing masses in same plane and in different planes. Cams: Types of Cams, Types of Followers. Displacement, Velocity & Acceleration Time Curves for Cam Profiles. Disc Cam with Reciprocating Follower Having Knife- Edge, Roller & Flat-Face Follower.	11
3	Disc Cam with Oscillating Roller Follower. Follower Motions including, SHM, Uniform Velocity, Uniform Acceleration & Retardation and Cycloidal Motion. Introduction to the design process, Factors influencing machine design, Selection of materials based on mechanical properties, design considerations, causes of failure in shafts, Direct, bending and torsional stress equations. Design of solid and hollow shafts based on strength, Rigidity and critical speed, material, design for static and fatigue loads, repeated loading, reversed bending.	11
4	Design of helical, leaf, disc and torsional springs under constant loads. Sliding contact bearing: - lubrication, lubricants, viscosity, journal bearings, hydrodynamic theory, Petroff's equation, bearing characteristic number, Sommerfeld number, Heat generated in bearings, Heat dissipated by bearings, Design procedure of Journal bearings. Ball and roller bearings: - Types, bearing life, static and dynamic load capacity, Stribeck's Equation, selection of bearings, selection of taper roller bearings, Design procedure of Ball and roller bearings, Needle bearings.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse the velocity and acceleration of the links and calculate the static and dynamic forces acting on them	K3
CO2	Understand and analyse various power transmission mechanisms including belts and cam & follower	K2
CO3	Calculate the masses required for balancing several rotating masses in same plane and in different planes.	K2
CO4	Explain the selection of material and design of solid and hollow shafts based on strength	K3
CO5	Be knowledgeable on the design of bearings and springs	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1			1						2
CO2	2	2			2							2
CO3	3	2										2
CO4	1	1	2			1						2
CO5	1	1	1			1						2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory of Machines	Rattan S.S	Tata McGraw-Hill Publishing Company Ltd., New ^[1] _{SEP} Delhi	3 rd - 2009
2	A Text Book of Machine Design	R. S. Khurmi and J. K. Gupta	Eurasia Publishing House, New Delhi	25 th - 2014
3	Mechanical Engineering Design	Joseph, E. Shigley and Charles R. Mischke	McGraw Hill International	6 th – 2003
4	Theory of Machines	Sadhu Singh	Pearson Education (Singapore) Pvt. Ltd, Indian Branch New Delhi	2 nd - 2006
5	Theory of Machines & Mechanisms	J.J. Uicker, G.R. Pennock, J.E. Shigley	OXFORD	3 rd - 2009
6	Theory of Machines	Thomas Bevan	CBS Publication	1984
7	Theory of Machines and Mechanisms	Ballaney P. L.	Khanna Publishers	2005

SEMESTER S4

SOIL & WATER CONSERVATION ENGINEERING

Course Code	PBAGT404	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	4	ESE Marks	40
Credits	3:0:0:1	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the causes and types of water erosion of soil and its estimation.
2. To understand agronomical and engg measures of controlling soil erosion
3. To conceptualise the process of gully erosion
4. To understand the functional design of various gully control structures
5. To learn structural design of various gully control structures.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Soil erosion - Introduction, causes and types - geological and accelerated erosion, agents, factors affecting and effects of erosion. Water erosion - Mechanics and forms - splash, sheet, rill, gully, ravine and stream bank erosion. Gullies - Classification, stages of development. Soil loss estimation – Universal soil loss equation (USLE) and modified USLE. Rainfall erosivity - estimation by KE>25 and EI30 methods. Soil erodibility - topography, crop management and conservation practice factors. Measurement of soil erosion - Runoff plots, soil samplers. Land capability classification. Rate of sedimentation, silt monitoring and storage loss in tanks.	11

2	Wind erosion- Factors affecting, mechanics, soil loss estimation and control measures - vegetative, mechanical measures, wind breaks and shelter belts and stabilization of sand dunes Water erosion control measures - agronomical measures – contour farming, strip cropping, conservation tillage and mulching. Engineering measures– Bunds and terraces. Bunds - contour and graded bunds – design. Terraces - level and graded broad base terraces, bench terraces - planning, design and layout procedure, contour stonewall and trenching.	11
3	Gully and ravine reclamation - principles of gully control - vegetative measures, temporary structures and diversion drains. Grassed waterways and design. Use of Geotextiles in soil and water conservation. Permanent structures for soil conservation and gully control - check dams, drop, chute and drop inlet spillways - design requirements, planning for design, design procedures - hydrologic, hydraulic and structural design and stability analysis. Drop spillway - applicability, types – straight drop, box-type inlet spillways - description, functional use, advantages and disadvantages, straight apron and stilling basin outlet, structural components and functions.	11
4	Hydraulic jump and its application. Loads on head wall, variables affecting equivalent fluid pressure, triangular load diagram for various flow conditions, creep line theory, uplift pressure estimation, safety against sliding, overturning, crushing and tension. Chute spillway - description, components, energy dissipaters, design criteria of Saint Antony Falls (SAF) stilling basin and its limitations. Drop inlet spillway - description, functional use and design criteria. Design of Diversions.	11

Suggestion on Project Topics

Course Assessment Method

(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the causes and types of water erosion of soil and its estimation.	K1
CO2	Understand agronomical and engg measures of controlling soil erosion	K2
CO3	Conceptualise the process of gully erosion	K3
CO4	Understand the functional design of various gully control structures	K2
CO5	Learn structural design of various gully control structures.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		1						1		1
CO2	3	2	2	1						1	2	1
CO3	3	1	2	2						1	2	1
CO4	3	2	2	1						1		
CO5	3	2		1						1		

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Agricultural Engineering. Volume II	Michael, A.M. and T.P. Ojha.	Jain Brothers, New Delhi.	4th Edition, 2003.
2	Soil and Water Conservation Engineering.	Suresh, R.	Standard Publisher Distributors, New Delhi.	2014.
3	Land and Water Management Engineering.	Murthy, V.V.N.	Kalyani Publishers, New Delhi	4 th Edition, 2002
4	Introduction to Soil and Water Conservation Engineering.	Mal, B.C.	Kalyani Publishers. , New Delhi	2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Soil and Water Conservation and Watershed Management.	Mahnot, S.C	International Books and Periodicals Supply Service, New Delhi.	2014
2	Soil and Water Conservation Engineering	Frevert, R.K., G.O. Schwab, T.W. Edminster and K.K. Barnes.	John Wiley and Sons, New York.	4 th edition, 2009
3	Soil Conservation.	Norman Hudson.	Cornell University Press, Ithaka, New York, USA.	1985

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S4

REFRIGERATION, AIR CONDITIONING AND COLD STORAGE

Course Code	PEAGT411	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the basic principles of refrigeration system and air refrigeration cyc
2. To understand the vapour compression and vapour absorption refrigeration system
3. To learn the principle of air conditioning, their different types and functions
4. To apply the design principles of cold storages to different food resources
5. To learn ice manufacturing principles and system of ice production

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Principles of refrigeration - Definition, background with second law of thermodynamics, unit of refrigerating capacity, coefficient of performance. Production of low temperatures- expansion of a liquid with flashing, reversible/ irreversible adiabatic expansion of a gas/ real gas, thermoelectric cooling, adiabatic demagnetization. Heat engine, heat pump and refrigerator – working principle and COP. Air refrigeration cycles - open, closed. Air refrigerators working on reversed Carnot cycle, Air refrigerators working on Bell Coleman cycle, Reversed Brayton or Joule or Bell Coleman cycle, analysis of gas cycle, polytropic and multistage compression.	9

2	Vapour compression refrigeration system (VCRS) -mechanism, COP, P-V, T-S, P-H diagrams and types of vapour compression cycles. Components of mechanical refrigerator. Study of compressors, condensers, expansion devices and evaporators. Vapour absorption refrigeration system - simple and practical, calculations, maximum coefficient of performance of a heat operated refrigerating machine. Types of refrigeration systems- Electrolux, Lithium Bromide absorption, Absorption type solar and Steam jet refrigeration systems. Common refrigerants -classification, nomenclature, desirable properties of refrigerants- physical, chemical, safety, thermodynamic and economical.	9
3	Psychrometry - thermodynamic properties of moist air, perfect gas relationship for approximate calculation, adiabatic saturation process, wet bulb temperature and its measurement, psychrometric chart and its use, elementary psychrometric processes. Air conditioning- principles- types and functions of air conditioning, physiological principles in air conditioning, adiabatic mixing of two air streams, sensible heat factor (SHF), RSHF, GSHF, ERSHF, air distribution and duct design methods - humidifiers and dehumidifiers- cooling load calculations, types of air conditioners- applications, design of complete air-conditioning systems.	9
4	Cold storage - cold store, design of cold storage for different categories of food resources, size and shape, construction and material, insulation, vapour barriers, floors, frost-heave, interior finish and fitting, evaporators, automated cold stores, security of operations. Ice manufacture, principles and systems of ice production, Treatment of water for making ice, brines, freezing tanks, ice cans, air agitation, quality of ice. Refrigerated transport: Handling and distribution, cold chain, refrigerated product handling, order picking, refrigerated vans, refrigerated display- applications of refrigeration and air conditioning.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understanding the basic principles of refrigeration system and air refrigeration cycles	K2
CO2	Understand the vapour compression and vapour absorption refrigeration system	K2
CO3	Understanding the principle of air conditioning, their different types and functions	K2
CO4	Apply the design principles of cold storages for the preservation of different food resources	K3
CO5	Understanding ice manufacturing principles and system of ice production	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	2					2			2
CO2	3	3	2								2	
CO3	3	3	1						2		1	
CO4	3	3	1	1								
CO5	3	2	1						2			

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Refrigeration & Air Conditioning Technology	William C. Whitman, William M. Johnson, John A. Tomczyk and Eugene Silberstein	Delmar, Cengage Learning, NY, USA	6th Ed, 2009
2	A Textbook of Refrigeration and Air Conditioning	R S Khurmi and J K Gupta	S. Chand Publishing	Revised edition, 2006
3	Refrigeration and Air Conditioning	W.F. Stoecker and J.W. Jones	McGraw-Hill Book Co., New York, USA	2nd Ed, 1982
4	Refrigeration and Air Conditioning	C.P. Arora	Tata McGraw-Hill Publishing Co. Ltd., New Delhi	2nd Ed, 2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Thermodynamics	Nag P K	Tata McGraw Hill Publishing Co. Ltd, New Delhi	
2	Thermal Engineering	Ballney P. L	Khanna Publishers, New Delhi	
3	Thermodynamics and Heat Power Engineering	Mathur M L and Mehta F S	DhanpatRai and Sons, NaiSarak, New Delhi	
4	A Course in Thermodynamics and Heat Engines	Kothandaraman C P Khajuria P R and Arora S C	DhanpetRai and Sons, NaiSarak, New Delhi	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/107/112107208/
2	https://archive.nptel.ac.in/courses/112/107/112107208/
3	https://archive.nptel.ac.in/courses/112/107/112107208/
4	https://archive.nptel.ac.in/courses/112/107/112107208/

SEMESTER S4

STORAGE ENGINEERING

Course Code	PEAGT 412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	0	Exam Hours	2 Hrs 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand about the basics of food grain storage and its design
2. To gain insights into CAS & MAS of fruits and vegetables
3. To understand the basics of precooling and cooling load calculations for different commodities

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Grain storage principles- factors- changes during storage. indoor storage system- Bag storage, bulk storage, bunkers, cap storage bags, outdoor storage system – cold storage, hermetic storage, vacuum and gas storage - moisture movement during bulk storage of grains – pressure distribution in storage bins - grain storage structures - location and material selection for storage building - types - traditional, modern, temporary and permanent storage structures, factors affecting qualities of grains during packaging and storage, functional requirements of storage structures - traditional storage structures in India and their improvements.	9

2	Grain storage ecosystem- biotic and abiotic factors- Moisture and temperature migration- Cold spots and hot spots. Aeration – Cooling- dehydration in grain storage, Grain storage in silos, bins and godowns – RCC and steel structures - aeration system for various storage structures, grain pressure theories - design of bulk storage structures, bag storage, godowns, Theory and nature of grain flow, pressure distribution, flow pattern shoppers and ducts – design - Loading and unloading mechanism, management and maintenance of grain storage, code of practices for safe storage of food grains.	9
3	Harvest and pre-harvest factors, pre-storage treatments, Controlled and modified atmosphere storage - principles, methods - optimization of storage gas composition, rate of supply, control systems for O ₂ and CO ₂ , effect of nitrogen, oxygen, and carbon dioxide on durable and perishable commodities, Effect of storage temperature on shelf life in CAS and MAS, Storage of Intermediate moisture Foods -storage of dehydrated fruits and vegetables.	9
4	Precooling methods of fruits & vegetables- Ventilation, Forced air cooling, Hydro-cooling, Evaporative cooling, Vacuum cooling, Cold storage of fruits & vegetables – Preliminaries for estimation of cooling load for optimum shelf-life of fruits & vegetables. Design of small capacity cold storage- Dimensions, Layout, stacking methods for cold storage, Transportation of fruits & vegetables – Types, stacking patterns, guidelines for best use of refrigerated transport. Design of cold storage for food products– Meat product, Fishery Product, Dairy Products.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

\End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the principles of grain storage	K1
CO2	Explain the storage system for cereal grains & pulses	K2
CO3	Apply the principles of CAS & MAS for extending the shelf-life of foods	K3
CO4	Utilize the precooling techniques and estimate cooling load for various commodities	K3
CO5	Design cold storage and transportation system for various commodities	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	1	1				1	1	2
CO2	3	3	3	1	1					1	1	2
CO3	3	3	3	2	1	2				1	1	2
CO4	3	3	3	2	2					1	1	2
CO5	3	3	3	2	1	1				1	1	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Post-Harvest Technology of Cereals, Pulses and Oilseeds	Chakravarty	IBH Publications	Third Edition, 2005
2	Principles of Agricultural Engineering	Ojha TP and Michael A.M	Jain Brothers	Tenth Edition, 2013
3	Modified and Controlled Atmospheres for the Storage, Transportation, and Packaging of Horticultural Commodities	Elhadi M. Yahia	Taylor & Francis Group	First Edition, 2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cooling Technology in the Food Industry	A. Ciobanu and G. Lasku, V. Bersescu	Abacus Press	First Edition, 1998
2	Preservation and storage of grains, seeds and their by – products	Multon, J.L	CBS Publishers and Distributors	Second Edition, 2000

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP_{fc} from GDP_{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP_{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, $NFIA=(-200)$ and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism , Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution -Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places -accessibility and social impacts, Managing conflict , Collective bargaining, Confidentiality , Role of confidentiality in moral integrity, Codes of Ethics . Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education , employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in	6

	connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental	6

	policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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**Course Assessment Method
(CIE: 50 marks , ESE: 50)**

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.

- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4

FLUID MECHANICS, OPEN CHANNEL HYDRAULICS & HYDRAULIC MACHINES

Course Code	PCAGL407	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To impart practical knowledge of fluid properties, fluid flow, open-channel flow, hydraulics and hydraulic machinery.
2. To provide hands-on practical training on various experimental setups to reinforce conceptual knowledge in fluid properties, fluid flow, open-channel flow, hydraulics and hydraulic machinery.

Expt. No.	Experiments
1	Determination of specific gravity, density and viscosity of the given liquid
2	Determination of total pressure and centre of pressure on plane surface
3	Determination of metacentric height of a floating body
4	Verification of Bernoulli's equation
5	Determination of coefficient of discharge and calibration of Orifice meter/venturimeter/Nozzle meter
6	Determination of hydraulic coefficients of orifice/mouthpiece

7	Determination of coefficient of discharge of orifice under varying head
8	Determination of coefficient of discharge and calibration of Notches
9	Determination of Chezy's constant and Darcy's coefficient on pipe friction apparatus
10	Determination of minor losses in pipes
11	Determination of Chezy's constant and Darcy's coefficient on open channel flow apparatus
12	Velocity of flow using current meter.
13	Performance test on centrifugal pumps,
14	Performance test on positive displacement pumps.
15	Performance test on axial flow pumps.
16	Performance test on mixed flow pumps

Course Assessment Method

(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fluid properties, statics, kinematics and dynamic aspects of fluid flow	K2
CO2	Calibrate all flow-measuring devices in Open channels and pipes	K3
CO3	Evaluate the major and minor losses in pipes for the design of irrigation systems	K5
CO4	Determine the efficiency and plot the characteristic curves of different types of pumps	K4
CO5	Select the most suitable pumps for different applications	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1								1
CO2	3	3	1	1								1
CO3	3	3	2	2								1
CO4	3	2	2	1	3							1
CO5	3	3	3	2	3							1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hydraulics & Fluid Mechanics, including hydraulic machines	Modi P. N. and S. M. Seth	S.B.H Publishers, New Delhi	18 th 2011
2	A Text Book of Hydraulics, Fluid Mechanics and Hydraulic Machines	Khurmi, R .S.	S. Chand & Company Limited, New Delhi	15 th , 1987
3	Fluid Mechanics: fundamentals and Applications	Cengel and Cimbala	McGraw Hill Education India	4th, 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Fluid Mechanics	Graebel. W. P,	Taylor & Francis	1 st , 2001
2	Fluid Mechanics with Engineering Applications	J. Frabzini	McGraw Hill	10 th , 2017
3	Fluid Mechanics: An Introduction	Radhakrishnan, E	Prentice Hall India	4 th , 2019
4	Mechanics of Fluids	Shames I. H	McGraw Hill	3rd 2013

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://r.search.yahoo.com/_ylt=AwrX_U1Szmdmy0URpkq7HAX.;_ylu=Y29sbwNzZzMEcG9zAzMEdnRpZAMEc2VjA3Ny/RV=2/RE=1718107858/RO=10/RU=https%3a%2f%2fme.iitp.ac.in%2fVirtual-Fluid-Laboratory%2f/RK=2/RS=UtANcem08IxUftSPRs7w71cRE-
2	https://r.search.yahoo.com/_ylt=AwrKGJFKz2dmzLQQpj7HAX.;_ylu=Y29sbwNzZzMEcG9zAzMEdnRpZAMEc2VjA3Ny/RV=2/RE=1718108107/RO=10/RU=https%3a%2f%2ffim-nitk.vlabs.ac.in%2f/RK=2/RS=xX47MvMdMLtwNbohMlvkuuV8uRw-
3	https://r.search.yahoo.com/_ylt=AwrKGJFKz2dmzLQQrDq7HAX.;_ylu=Y29sbwNzZzMEcG9zAzYEEdnRpZAMEc2VjA3Ny/RV=2/RE=1718108107/RO=10/RU=https%3a%2f%2feerc03-iiith.vlabs.ac.in%2fexp%2fventuri-meter%2f/RK=2/RS=2egvsBn5QVkBjOQZ1HkcesmZTzU-
4	https://r.search.yahoo.com/_ylt=AwrKBYfP0GdmqWMJSCG7HAX.;_ylu=Y29sbwNzZzMEcG9zAzEEEdnRpZAMEc2VjA3Ny/RV=2/RE=1718108495/RO=10/RU=https%3a%2f%2fwww.vlab.co.in%2finstitute_detail.php%3fins%3d011/RK=2/RS=U4kjbVf27F9APAk3ZKv8KoXA5I-

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.

- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

1. Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S4

SOIL SCIENCE & MECHANICS LAB

Course Code	PCAGL408	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course objectives:

1. Students will learn the basic and engineering behaviour of soils.
2. They will understand the basic properties of soil and different laboratory methods for determination of these basic properties of soil.

Expt. No.	Experiments
1	Determination of Field Capacity, PWP and Water holding capacity of soil
2	Determination of pH and EC of soil
3	Determination of field density of soil using core cutter method
4	Determination of field density using sand replacement method
5	Particle size distribution by sieve analysis
6	Specific gravity and water content by pycnometer
7	Liquid limit, plastic limit
8	Sedimentation analysis
9	Compaction test
10	Shrinkage limit
11	Constant head test to determine coefficient of permeability
12	Falling head test ,,,

13	Determination of shear parameters by direct shear test
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Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understanding of the basic properties of soil	K1
CO2	Perform laboratory methods for determination of the basic properties of soil	K3
CO3	Understanding of the Engineering behaviour of the soil	K1
CO4	Perform different laboratory methods for determination of the engineering properties of soil.	K3
CO5	Understanding of the basic application of the concepts of the soil mechanics	K1

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1								1
CO2	3	3	1	1								1
CO3	3	3	2	2								1
CO4	3	2	2	1	3							1
CO5	3	3	3	2	3							1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Text Book of Soil Mechanics and Foundation Engineering	V.N.S. Murthy	Saikripa Technical consultants, Bangalore	2 nd edition, 2011
2	Geotechnical Engineering	S. K. Gulati et. al.	TMH Publication Co. Ltd., New Delhi	2021
3	Basic and applied Soil Mechanics	Gopal Ranjan and A. S. R. Rao	Wiley Eastern Ltd., New Delhi	2015
4	Soil Mechanics and Foundation Engineering	K. R. Arora	Standard Pub. and Dist., Delhi	2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Fluid Mechanics	Graebel. W. P,	Taylor & Francis	1 st , 2001
2	Fluid Mechanics with Engineering Applications	J. Frabzini	McGraw Hill	10 th , 2017
3	Fluid Mechanics: An Introduction	Radhakrishnan, E	Prentice Hall India	4 th , 2019
4	Mechanics of Fluids	Shames I. H	McGraw Hill	3rd 2013

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://r.search.yahoo.com/_ylt=Awrx_U1Szmdmy0URpkq7HAX.;_ylu=Y29sbwNzZzMEcG9zAzMEdnRpZAMEc2VjA3Ny/RV=2/RE=1718107858/RO=10/RU=https%3a%2f%2fme.iitp.ac.in%2fVirtual-Fluid-Laboratory%2f/RK=2/RS=UtANcem08IxUftSPRsfh7w71cRE-
2	https://r.search.yahoo.com/_ylt=AwrKGJFKz2dmzLQQpj7HAX.;_ylu=Y29sbwNzZzMEcG9zAzMEdnRpZAMEc2VjA3Ny/RV=2/RE=1718108107/RO=10/RU=https%3a%2f%2ffim-nitk.vlabs.ac.in%2f/RK=2/RS=xX47MvMdMLtwNbohMIvkuuV8uRw-
3	https://r.search.yahoo.com/_ylt=AwrKGJFKz2dmzLQQrDq7HAX.;_ylu=Y29sbwNzZzMEcG9zAzYEdnRpZAMEc2VjA3Ny/RV=2/RE=1718108107/RO=10/RU=https%3a%2f%2feerc03-iiith.vlabs.ac.in%2fexp%2fventuri-meter%2f/RK=2/RS=2egvsBn5QVkBjOQZ1HkcesmZTzU-
4	https://r.search.yahoo.com/_ylt=AwrKBYfP0GdmqWMJSCG7HAX.;_ylu=Y29sbwNzZzMEcG9zAzEEdnRpZAMEc2VjA3Ny/RV=2/RE=1718108495/RO=10/RU=https%3a%2f%2fwww.vlab.co.in%2finstitute_detail.php%3fins%3d011/RK=2/RS=U4kjbVvf27F9APAk3ZKv8KoXA5I-

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

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- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

AGRICULTURE ENGINEERING

SEMESTER S5
DESIGN OF FARM STRUCTURES AND COST ESTIMATION

Course Code	PCAGT501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To acquire knowledge and skills in designing reinforced concrete structures, including beams, columns, foundations, and connections.
2. To study the different types of structural systems (such as beams, columns, slabs, frames, and trusses) and their applications in building and infrastructure projects.
3. To understand the design of simple steel structures
4. To learn the principles of designing farm structures, including considerations such as structural stability, environmental factors (like wind and snow loads), ventilation, lighting, and animal comfort.
5. To develop skills in estimating the costs associated with designing and constructing farm structures, considering factors such as materials, labour, equipment, and overhead costs.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Design of a Slab- Design of slab spanning in one direction and two directions. Columns- Design of short columns with axial loads, Design of long column (Brief description only).	11

	Footings- Design of Isolated footings- axial and eccentric loading- Design of Combined footings	
2	Design of steel structures – Concept of design of steel structures, Use of IS codes and Hand books, Properties of structural steel. Design of connections- types of connection, design of welded connection and riveted connection Design of tension members and compression members, Beam-Design of laterally supported and unsupported beams Columns- Design of axially loaded compression members, lacing and battening, Column bases- slab bases.	11
3	Design of farm infrastructures- functional and structural design of Farm infrastructures and analysis of stability of the structures-design of residential buildings, farm offices, Poly houses, open wells, farm roads, culverts, Sewage system and its design. Design of septic tank for small family – Soak pit. Functional design and layout of a dairy farm, poultry house, goat house, drying yards, Design of a feed/fodder storage structures, Design of grain storage structures, Design and layout of commercial bag and bulk storage facilities, Performance evaluation of different domestic storage structure. Design of a farm fencing system, .	11
4	Estimation, basic terms and types of estimates - revised estimate, supplementary estimate, maintenance estimate, approximate estimate. Methods of estimate preparation- plinth area method, cubic rate method, unit rate method, bay method, approximate quantity from bill method, comparison method, cost from materials and labour. Preparation of detailed estimate for buildings - centre line and ‘long wall - short wall’ method. Preparation of detailed estimate for sanitary and water supply works, road, irrigation works, steel structures, doors and windows. Preparation of detailed estimate for RCC Structures, Preparation of bar bending schedule. Detailed specifications for common building materials and items of work as per I.S specifications, preparation of conveyance statement, calculation of	11

	quantities of materials, analysis of rates and preparation of abstract of estimate of agricultural engineering works, Measurement books, Use of PRICE software. Valuation- Purpose and principles of valuation, Definition of various terms related to valuation - depreciation, obsolescence, sinking fund, obsolescence, salvage and scrap value, market value, fair rent, year's purchase etc. Methods of valuation - Replacement cost method, Rental return method, Profit based method and Depreciation method. Valuation of land - belting method, development method, and hypothecated building scheme method. Rent calculation and lease.	
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Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recall and remember the fundamental properties of different building materials and structural elements.	K1
CO2	Design reinforced concrete and steel structures, including beams, columns, slabs, foundations, and connections, considering structural integrity, durability, and economic feasibility.	K3
CO3	Understand the purposes, functions, and layout considerations of farm structures in various agricultural operations.	K2
CO4	Apply design codes and standards appropriately to ensure structural designs meet safety, performance, and regulatory requirements.	K3
CO5	Develop skills in cost analysis and budgeting for agricultural projects involving construction or renovation of farm facilities.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1											2
CO2	3	2	2	1	2	3	1	3	3		2	
CO3		2	2			2	1	2	1	2		2
CO4	2	2	3	2		3	1	3	3		2	
CO5	1		2		2	1	1	2			3	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of Steel Structures Vol. I & II.	Ramachandra and V. Gehlot	Scientific Publishers	2015
2	Design of Reinforced Concrete Structures	Ramamrutham, S. and Narayan, R.	Dhanpat Rai Publishing Co Pvt Ltd.	2016
3	Principles of Agricultural Engineering, Vol. I.	Ojha, T.P and Michael, A.M.	Jain Brothers, Karol Bag, New Delhi	2002
4	Estimating and Costing in Civil Engineering,	Dutta, B.N.	Duttta & CO, Lucknow	1969

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of Reinforced Concrete Structures	Vazirani, V.N. and Ratwani, M.M.	KhannaPublishers	2013
2	Design Of Steel Structures	Ramamrutham, S.	Dhanpat Rai Publishing Co Pvt Ltd.	1986
3	Principles and practices of Agricultural Structures and Environmental Control	Pandey, P.H.	Kalyani Publishers, Ludhiana	2004
4	Estimating, Costing, Specification and Valuation in Civil Engineering	Chakraborthi, M.	M Chakraborti	2010

Video Links (NPTEL, SWAYAM...)

Module No.	Link ID
1	https://www.youtube.com/watch?v=3Do2cZMUGYc&list=PL51300B0778FB5784&index=2
2	https://youtu.be/LrDdQvXnv-0
3	http://ecoursesonline.iasri.res.in/course/view.php?id=17
4	https://www.youtube.com/watch?v=WcfBJ8H3kkg&list=PLQibxotQ9iEgCX1pgE3HW62rbU5fsreBH

SEMESTER S5
THERMODYNAMICS AND HEAT ENGINES

Course Code	PCAGT502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Engineering Mathematics Physical Chemistry	Course Type	Theory

Course Objectives:

1. Impart knowledge on the basics and application of zeroth, first and second law of thermodynamics.
2. To learn various properties of steam through steam tables and Mollier chart.
3. To analyze the working of IC engines and various auxiliary systems present in IC engines
4. To learn the concepts and laws of thermodynamics to predict the operation of thermodynamic cycles

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	BASICS AND LAWS OF THERMODYNAMICS Review of Basics – Thermodynamic systems, Properties and processes Thermodynamic Equilibrium - Displacement work - p-V diagram. Thermal equilibrium - Zeroth law – Concept of temperature and Temperature Scales. First law – application to closed and open systems – steady flow processes.	11

	Heat Engine – Refrigerator - Heat pump. Statements of second law and their equivalence & corollaries. Second law of thermodynamics: Carnot cycle - Reversed Carnot cycle - Performance - Clausius inequality.– Principle of increase of Entropy	
2	PROPERTIES OF PURE SUBSTANCES Steam - formation and its thermodynamic properties - p-V, p-T, T-V, T-S, h-S diagrams. p-V-T surface. Determination of dryness fraction. Calculation of work done and heat transfer in non-flow and flow processes using Steam Table and Mollier Chart.	11
3	INTERNAL COMBUSTION ENGINES IC engine – Classification, working, components and their functions. Ideal and actual : Valve and port timing diagrams, p-V diagrams- two stroke & four stroke, and SI & CI engines – comparison. Fuel properties - Fuel supply system- study of a simple carburettor-Study of fuel injection system of diesel engine – fuel pump and injectors- Lubrication system lubricant properties and classification. Cooling system-study Cooling system-Air cooling and water/liquid cooling systems. Ignition system- Battery Ignition and Magneto Ignition systems.	11
4	GAS AND STEAM POWER CYCLES Air Standard Cycles - Otto, Diesel, Dual, Brayton – Cycle Analysis, Performance and Comparison – Rankine, reheat and regenerative cycle	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the laws of thermodynamics in closed and open engineering systems and analyse the performance of thermal devices through energy and entropy calculation.	K1
CO2	Apply the properties of pure substance in computing the macroscopic properties of ideal and real gases using gas laws and appropriate thermodynamic relations.	K3
CO3	Explain the features and functioning of IC engines, their components and auxiliaries	K2
CO4	Apply thermodynamic concepts to different air standard cycles and solve problems	K3
CO5	Apply thermodynamic concepts to steam power cycles and solve problems	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	1	-	-	-	-	-	-	-	2
CO2	3	3	2	1	-	1	-	-	2	-	1	2
CO3	3	2	1	1	-	-	-	-	-	-	-	1
CO4	3	2	1	1	-	-	-	-	-	-	-	1
CO5	3	2	1	1	-	-	-	-	-	-	-	1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A course in thermal Engineering	Kothandaraman.C.P., Domkundwar. S,Domkundwar. A.V	Dhanpat Rai & sons	Fifth Edition 2016
2	Thermal Engineering	Rajput. R. K	S.Chand Publishers	2017
3	Engineering Thermodynamics	Nag.P.K.	Tata McGraw Hill, New Delh	6th Edition, 2017
4	Internal Combustion Engines	Ganesan V	Tata McGraw Hill, New Delh	6 th Edition, 2012
5	Refrigeration and Air Conditioning ,	Arora.C.P,	Tata McGraw-Hill Publishers	2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Engineering Thermodynamics: Fundamentals and Applications”	Natarajan, E.,	Anuragam Publications, Chennai	2 nd Edition, 2014
2	Thermodynamics - An Engineering Approach	Cengel, Y and M. Boles, ,.	Tata McGraw Hill	,9th Edition, 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/105/112105123/
2	https://archive.nptel.ac.in/courses/112/105/112105266/
3	https://archive.nptel.ac.in/courses/112/103/112103262/
4	https://onlinecourses.nptel.ac.in/noc21_me21/preview#:~:text=The%20course%20contains%20the%20analysis,for%20power%20generation%20and%20condensers.

SEMESTER S5
HEAT AND MASS TRANSFER

Course Code	PCAGT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Engineering Chemistry Engineering Mathematics	Course Type	Theory

Course Objectives:

1. To understand the physical behavior of the various modes of heat transfers
2. To understand the concepts of heat transfer through extended surfaces.
3. To develop understanding about the methods of determination of heat transfer rates in conduction, convection and radiation.
4. To learn the thermal analysis and sizing of heat exchangers
5. To understand the basic concepts of mass transfer and its applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	CONDUCTION Basic concepts – Mechanism of heat transfer – Conduction, convection and radiation – Thermal properties of materials and measurement - General differential equation of heat conduction – Fourier law of conduction – Cartesian and cylindrical coordinates – one dimensional steady state heat conduction – Conduction through plane walls, cylinders and spherical systems – Composite systems – Conduction with internal heat generation – Extended surfaces	9

2	CONVECTION Basic concepts – Convective heat transfer coefficients – Boundary Layer concept – Types of convection – Forced convection – Dimensional analysis – Useful non dimensional numbers– External flow – Flow over plates, Cylinders and spheres – Internal flow – Laminar and turbulent flow – Combined Laminar and turbulent flow – Flow over bank of tubes – Free convection - Dimensional analysis – Flow over vertical plates, horizontal plate, inclined plate, cylinders and spheres.	9
3	RADIATION Basic concepts of radiation, Absorptivity, Reflectivity and Transmittivity of radiation,law of radiation – Stefan Boltzmann law, Kirchoff law – Black body radiation – Grey body radiation - shape factor algebra – solid angle, intensity of radiation. Radiation exchange between black surfaces, geometric configuration factor, Electrical analogy – Radiation shields – introduction to gas radiation	9
4	HEAT EXCHANGER AND MASS TRANSFER Types of heat exchangers – LMTD method of heat exchanger analysis – Overall heat transfer coefficient – Fouling Factors-Heat exchanger performance- transfer units- Basic concepts – Diffusion mass transfer – Fick’s Law of diffusion – Steady state molecular diffusion – Convective mass transfer – Momentum, heat and mass transfer analogy – Convective mass transfer correlations.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the physical behavior of the various modes of heat transfers	K1
CO2	Understand the concepts of heat transfer through extended surfaces.	K2
CO3	Learn the methods of determination of heat transfer rates in conduction, convection and radiation.	K2
CO4	Learn the thermal analysis and sizing of heat exchangers	K2
CO5	Explain the basic concepts of mass transfer and its applications	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	2	1	-	-	-	-	-	-	-	2
CO2	3	2	2	1	2	-	1	-	-	-	-	-
CO3	3	1	2	1	1	-	2	-	-	-	-	2
CO4	3	2	2	2	2	-	1	-	-	-	-	-
CO5	3	-	1	-	-	-	2	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Engineering Heat and Mass Transfer	Sachdeva, R.C	New Age International, New Delhi	1995
2	Heat Transfer	Nag, P.K.,	Tata McGraw Hill Book Co., New Delhi,	2002
3	Heat and Mass Transfer	R. K. Rajput	S Chand and Company Pvt. Ltd.	2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Heat and Mass Transfer	Yadav, R	Central Publishing House, New Delhi	1995
2	Heat and Mass transfer	Holman, J.P.	Tata McGraw Hill Book Co., New York	2002
3	Fundamentals of Engineering Heat and Mass Transfer	Kothandaraman, C.P	New Age International, New Delhi,	1998
4	Heat & Mass Transfer	Velraj, R.,	Ane Books, New Delhi,	2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/105/112105271/
2	https://archive.nptel.ac.in/courses/112/105/112105271/
3	https://archive.nptel.ac.in/courses/112/107/112107256/
4	https://archive.nptel.ac.in/courses/112/105/112105248/ https://archive.nptel.ac.in/courses/103/103/103103145/

SEMESTER S5
IRRIGATION AND DRAINAGE ENGINEERING

Course Code	PBAGT504	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. To acquaint and equip the students with the basic principles of irrigation and Soil-Plant Water relations
2. Understand different kinds of irrigation systems and water control and diversion structures and to introduce the concept of command area development
3. To introduce and impart the design concepts of micro-irrigation and Sprinkler irrigation systems.
4. Understand the basic concepts for planning, design and management of land drainage works in cultivated areas

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	WATER RESOURCES AND IRRIGATION REQUIREMENT Water Resources- Introduction to irrigation; land grading and levelling- estimation of earth work; Soil water plant relationship, soil water movement, infiltration; Determination of evapotranspiration-Crop water requirement, irrigation requirement, irrigation scheduling and irrigation efficiencies; Introduction to different types of irrigation.	9
2	METHODS OF IRRIGATION AND COMMAND AREA DEVELOPMENT	

	Surface and Subsurface methods of Irrigation-Design of lined and unlined canals-Silt theories- Kennedy's and Lacey's theories, Lining of water courses and field channels; Underground pipeline irrigation system design. Water conveyance -Water control and diversion structures; Command area - Concept, Components of CADP, Classification of canals- Alignment of canals.	9
3	MICRO IRRIGATION AND SPRINKLER IRRIGATION Micro Irrigation, components -Drip, micro sprinkler, mist, fogger etc.; Design of drip irrigation system; Maintenance of micro irrigation systems; Sprinkler irrigation, types of sprinkler irrigation systems; design of sprinkler irrigation system: layout selection. Performance evaluation of sprinkler irrigation system: uniformity coefficient and pattern efficiency; – Determination of pump capacity from head and discharge, selection and installation of pumps.	9
4	AGRICULTURAL DRAINAGE Agricultural drainage –Factors affecting drainage; Drainable porosity, Drainage coefficient; Principles of flow through soils, Darcy's law; Surface drainage systems - Subsurface drainage - Design of subsurface drainage - Steady state equations- Pipe materials - mole drains, drainage wells, Leaching requirements - irrigation and drainage water quality.	9

Suggestion on Project Topics

Course Assessment Method (CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the Planning, design, operation and management of Water Resources and Irrigation Requirement.	K1
CO2	Acquire knowledge on different methods of irrigation including canal irrigation.	K3
CO3	Understand different types Diversion and Impounding Structures	K2
CO4	Understand the concept Canal Irrigation and Command Area Development	K2
CO5	Understand the need and methods of agricultural drainage.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	2	2	2	1	2	1	2	2	2
CO2	2	2	3	2	2	1	3	2	2	2	2	2
CO3	2	2	2	2	2	3	1	1	1	2	2	2
CO4	3	3	2	3	2	1	3	1	1	2	2	2
CO5	3	3	3	3	2	1	1	2	2	3	2	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Irrigation Theory and Practice	Michael, A.M	Vikas Publishing House Pvt Ltd., Delhi.	2008
2	Irrigation Engineering	Garg S K	Laxmi Publication	2008
3	Irrigation Water Management	Dilip Kumar Majumdar	Prentice-Hall of India, New Delhi	2008
4	Drainage Principles and Applications	Ritzema, H.P	Publication No. 16, International Institute of Land Reclamation and Improvement, Netherlands	1994

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Irrigation Engineering	Basak, N.N	Tata McGraw-Hill Publishing Co, New Delhi	2008
2	Land and water management	Murthy, V.V.N	Kalyani publishing, New Delhi	1998
3	Land Drainage – Principles, Methods and Applications	Bhattacharya, A.K., and Michael, A.M	Konark Publishers Pvt. Ltd., New Delhi	2003
4	Drainage Principles and Applications	Kessler, J	Vol. II and IV, International Institute of Land Reclamation and Improvement, Netherlands	1979.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/126/105/126105010/
2	https://onlinecourses.nptel.ac.in/noc21_ag04/preview#:~:text=In%20this%20course%20we%20will,of%20agricultural%20lands%2C%20management%20of
3	https://onlinecourses.nptel.ac.in/noc20_ag04/preview
4	https://onlinecourses.nptel.ac.in/noc21_ag04/preview#:~:text=In%20this%20course%20we%20will,of%20agricultural%20lands%2C%20management%20of

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S5

REMOTE SENSING AND GIS

Course Code	PEAGT521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Hydrology and Water Resources Engineering, Soil Science	Course Type	Theory

Course Objectives:

1. To introduce the principles and basic concepts of Remote Sensing and GIS
2. To introduce the remote sensing systems, data products and analysis
3. To introduce concepts of GIS and map projections
4. To study the applications of Remote Sensing and GIS in agriculture, soil and water resources

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	CONCEPTS OF REMOTE SENSING AND SATELLITES Definition- Historical background - Components of remote sensing – Energy source, electromagnetic spectrum, radiation laws, platforms and sensors - Active and passive remote sensing satellites - Types - Sun synchronous - Geo synchronous remote sensing satellites - Resolutions - Spectral, spatial, radiometric and Temporal resolution- LANDSAT, Sentinel and IRS – Data acquisition - Concept of spectral radiance, Digital Number and Reflectance- Energy interaction with earth surface feature-Atmospheric effects on remote	9

	sensing –Spectral reflectance curves-spectral signatures for water, soil and vegetation- – Vegetation Indices	
2	DATA PRODUCTS AND IMAGE ANALYSIS Data products –based on level of processing- o/p – scale – area/coverage – data availability - Image interpretation – Visual interpretation elements – interpretation key. Digital image processing – Image enhancement – image classification – Supervised and unsupervised.	9
3	CONCEPTS OF GIS History and development of GIS – Definition – Components of GIS – Characteristics of Maps – Elements – Map scale, Vector and Raster data models- Attribute data -Map Projections, Coordinate systems – Sources of spatial Data-Spatial analysis in GIS-Spatial interpolation using GIS.	9
4	APPLICATION OF RS AND GIS Crop Acreage estimation - Estimation of Crop Water Requirement using thermal remote sensing – Crop condition - Soil mapping – classification of soil with digital numbers – soil erosion mapping- reservoir sedimentation using image processing - Inventory of water resources – water quality assessment - Application of Remote Sensing and GIS in Precision Agriculture - Monitor Crop Health - Management Decision Support Systems	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the remote sensing principles and systems.	K1
CO2	Gain sufficient knowledge on satellite data processing and available data products	K2
CO3	Know the concept of GIS and its tools.	K2
CO4	Have knowledge on data input and analysis techniques.	K2
CO5	Utilize these advanced techniques in addressing the real world problems	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	-	1	-	1	1	1	2	3
CO2	3	2	3	2	1	-	2	1	-	1	2	3
CO3	3	3	2	-	1	-	-	-	2	2	3	3
CO4	2	3	1	-	-	-	3	1	-	1	2	2
CO5	2	3	2	3	2	3	3	1	2	1	2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Remote Sensing and Geographical Information Systems	Anji Reddy. M	BS Publications, Hyderabad	2001
2	Remote Sensing and Image Interpretation	Lillesand, T. M., and Kiefer, R.W	John Wiley and Sons, New York	2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Geographical Information System: Applications in Forestry and Natural Resources Management	Bettinger, P., and Michael, G.W	Tata McGraw–Hill Higher Education, New Delhi	2003
2	An Introduction to GIS,	Ian Heywood	Pearson Education, New Delhi	2001
3	Geographical Information System – An Introduction	Jeffery Star and John Estes	Prentice Hall India Pvt. Ltd., New Delhi	1998
4	Remote sensing principles & applications	Patel A.N & Surendra Singh	Scientific Publishers , Jodhpur	1992
5	Introduction to Geographic Information Systems	Kang-Tsung Chang	McGraw Hill Education	2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/105/103/105103193/
2	https://onlinecourses.nptel.ac.in/noc19_ce38/preview
3	https://archive.nptel.ac.in/courses/105/107/105107155/
4	https://onlinecourses.nptel.ac.in/noc22_ce84/preview#:~:text=This%20course%20is%20design ed%20to,in%20their%20future%20research%20work.

SEMESTER S5
PUMP ENGINEERING AND HYDRAULIC CONTROL

Course Code	PEAGT522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To equip students with the necessary knowledge and skills essential for addressing the dynamic challenges in the agricultural sector.
2. To embark the students on a journey that spans from foundational principles to advanced concepts, preparing them for a future where sustainable and efficient water management in agriculture is paramount.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic Principles of Fluid Mechanics-Introduction to Fluid Mechanics- Fundamentals of fluid properties. Types of flow: laminar, turbulent, and transitional- Bernoulli's equation and its relevance to pump systems. Flow in Pipes- Pipe friction losses and head calculations - Reynolds number and its significance - Hydraulic gradients in pump systems. Cavitation: causes, effects, and prevention- Net Positive Suction Head (NPSH) and its importance. Introduction to Pump Systems- Importance of pump systems in agriculture-Types of pumping systems in agriculture- Types of Pumps and Their Applications- Centrifugal pumps: working principles and applications - Positive displacement pumps: reciprocating and rotary pumps- Specialized	9

	pumps for agricultural applications- Selection Criteria for Agricultural Pumps.	
2	Introduction to Hydraulic Control Systems - Definition and significance of hydraulic control in agricultural machinery- Overview of hydraulic components: pumps, actuators, valves, and reservoirs- Hydraulic Symbols and Schematics- Control Valves and Their Types-Classification of Hydraulic Control Valves- Flow Control in Hydraulic Systems-Pressure Regulation in Hydraulic Systems.	9
3	Pump Maintenance and Troubleshooting-Preventive Maintenance of Pumps and Hydraulic Systems - Preventive Maintenance Practices- Troubleshooting Common Issues in Pump Systems- Overhauling Hydraulic Components- Identifying and mitigating safety hazards in maintenance activities- Personal protective equipment (PPE) requirements.	9
4	Strategies for Energy-Efficient Pumping- Pump system optimization for energy efficiency - Energy audit techniques for pump systems - High-efficiency pump designs - Smart pump systems and their applications- Automation in Pump Control Systems- Programmable Logic Controllers (PLCs) in pump control- Remote sensing and monitoring of pump parameters- Sustainable pump technologies and eco-friendly materials- Anticipating and adapting to future developments.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of fluid mechanics.	K1
CO2	Identify different types of pumps, understand their working principles, and evaluate their performance characteristics.	K3
CO3	Design and analyse hydraulic control systems, integrating them into agricultural machinery for optimal performance.	K3
CO4	Perform routine preventive maintenance, troubleshoot common issues, and carry out repairs on pump systems used in agriculture.	K2
CO5	Critically analyse emerging trends and technologies in pump engineering.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		3	2								1
CO2	2	2	1	2	1		3					
CO3	3	3	2	3	1		2					1
CO4	3	2	3	2	2		2					
CO5	2		1				1					1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Variable Speed Pumping: A Guide to Successful Applications	Anonymous	Europump & Hydraulic Institute	2011
2	Fluid Mechanics	Frank M. White	McGraw Hill	2022
3	Pump Handbook	Igor J. Karassik, Joseph P. Messina, Paul Cooper, and Charles C. Heald	Oxford & IBH.	2001
4	Pump User's Handbook: Life Extension	Heinz P. Bloch and Allan R. Budris	Oxford & IBH	2012

SEMESTER S5
FUNDAMENTALS OF NANOSCIENCE

Course Code	PEAGT 523	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. This course is designed to provide a comprehensive understanding of organic semiconductors and action of various layers used in organic devices.
2. The students will gain the skills necessary to design and develop organic devices like OLEDs, OFETs and OPVs after the completion of the course.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of Organic semiconductor, conduction mechanism in organic semiconductors, Pi and Sigma electron band theory. Fundamentals of conducting polymers. Charge transport in conjugated polymers. Effect of solitons. Conduction mechanism in doped polymeric semiconductors. Luminescence, injection and transports properties in organic semiconductors.	9
2	Organic light emitting diodes(OLED) and Polymer light emitting diodes (PLED) – Basic device architecture, Single layer architecture and multilayer architecture, electrical properties and operating characteristics, Flexible display device architecture, Fabrication and characterization, Principle and architecture of organic transistors.	9
3	Basic principle and architecture of Plastic Solar cell. Charge transport, effects of excitation diffusion, dissociation and luminescence, Photo-generation	9

	process in organic heterojunction, photovoltaic cells. Processing of organic solar cells. Dye sensitization – dye sensitized solar cell.	
4	Essential characteristics and conductivity of electrode materials in organo electronic devices, Effect of Indium Tin Oxide (ITO) oxidative properties on efficiency and shelf life of organic electronic devices, Buffer organic layer protection to the active layer. Doping and for increased efficiency and shelf life– architecture. Devices fabrication- Thin film coating technique. Spin coating, dip coating, doctor blading screen printing, inkjet printing, vapor deposition. R.F and microwave plasma assisted film coating.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of charge transport, energy levels and doping in organic semiconductors and conducting polymers. Exemplify the properties and applications of organic electronic devices.	K1
CO2	Understand the device architecture of organic devices.	K2
CO3	Distinguish action of different layers of organic devices	K3
CO4	Define the different techniques employed to design and develop organic devices like OLEDs, OFETs and OPVs.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	3	2	1	1				1	2
CO2	3	3	3	3	2	1	1				1	2
CO3	3	3	3	3	2	1	1				1	2
CO4	3	3	3	3	2	1	1				1	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Supra molecular photo sensitive and electro–active materials	Nalwa,	Elsevier	2001
2	Organic Electronic Materials: Conjugated polymers and low molecular weight organic solid	R.Farchioni, G.Grosso	Springer series in materials science	2020
3	Organic electronics	Klausk Hagen	Wiley	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Flexible flat panel display	Gregory Crawford	Wiley	2000
2	Thin film materials for large area electronics	Equer	Elsevier	2010
3	Advanced synthetic metals	Bernier	Elsevier	2020

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/105/112105271/
2	https://archive.nptel.ac.in/courses/112/105/112105271/
3	https://archive.nptel.ac.in/courses/112/107/112107256/
4	https://archive.nptel.ac.in/courses/112/105/112105248/ https://archive.nptel.ac.in/courses/103/103/103103145/

SEMESTER S5

REFRIGERATION AND AIR CONDITIONING LABORATORY LAB

Course Code	PCAGL507	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To impart practical understanding of refrigeration and air conditioning system
2. Evaluate the performance of refrigeration and air-conditioning systems using various performance parameters.
3. To learn determination of refrigeration load
4. To understand the method of determination of cooling, humidification and dehumidification load of air conditioners.
5. Familiarize oneself with the latest advancements in the field of refrigeration

Expt. No.	Experiments
1	Tutorials on air refrigeration cycle systems
2	Study of domestic refrigerator
3	Studies on Heat Exchangers and their performance
4	Determination of COP of a refrigeration system
5	Estimation of refrigeration load for cold storage.
6	Study and tutorials on psychrometry
7	Determination of humidity of air using Sling Psychrometer.

8	Determination of capacity and COP of Split Air Conditioner
9	Performance studies on cascade refrigeration systems
10	Performance studies on heat pump for different indoor and outdoor conditions
11	Study of absorption type solar refrigeration system
12	Estimation of humidification and dehumidification load of air conditioner
13	Study and performance experiment on vapour absorption, refrigeration system.
14	Performance studies on vapour compression refrigeration system with air / water cooled condensers
15	Studies on Air Distribution Systems
16	Visit to Ice plant and cold storage or Building Energy Studies

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand refrigeration and air conditioning system	K1
CO2	Evaluate the performance of refrigeration and air-conditioning systems using various performance parameters.	K3
CO3	Determine of refrigeration load	K3
CO4	Determine cooling, humidification and dehumidification load of air conditioners.	K3
CO5	Familiarize with the latest advancements in the field of refrigeration.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	-	1	-	3	-	-	2	3	1	3	1
CO2	1	-	3	-	3	-	-	2	3	1	3	1
CO3	1	-	3	-	3	-	-	2	3	1	3	1
CO4	1	-	3	-	3	-	-	2	3	1	3	1
CO5	1	-	3	-	3	-	-	2	3	1	3	1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Refrigeration & Air Conditioning Technology	William C. Whitman, William M. Johnson, John A. Tomczyk and Eugene Silberstein	Delmar, Cengage Learning, NY, USA	6th Ed, 2009
2	A Textbook of Refrigeration	R S Khurmi and J K	S. Chand Publishing	Revised

	and Air Conditioning	Gupta		edition, 2006
3	Refrigeration and Air Conditioning	W.F. Stoecker and J.W. Jones	McGraw-Hill Book Co., New York, USA	2nd Ed, 1982
4	Refrigeration and Air Conditioning	C.P. Arora	Tata McGraw-Hill Publishing Co. Ltd., New Delhi	2nd Ed, 2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Thermodynamics	Nag P K	Tata McGraw Hill Publishing Co. Ltd, New Delhi	1995
2	Thermal Engineering	Ballney P. L	Khanna Publishers, New Delhi	1994
3	Thermodynamics and Heat Power Engineering	Mathur M L and Mehta F S	DhanpatRai and Sons,NaiSarak, New Delhi	1992
4	A Course in Thermodynamics and Heat Engines	Kothandaraman C P Khajuria P R and Arora S C	DhanpetRai and Sons, NaiSarak, New Delhi	1992

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/107/112107208/
2	https://archive.nptel.ac.in/courses/112/107/112107208/
3	https://archive.nptel.ac.in/courses/112/107/112107208/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S5

HEAT ENGINES LABORATORY LAB

Course Code	PCAGL508	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To understand the value timing-V diagram and performance of IC Engines.
2. To Study the characteristics of fuels/Lubricates used in IC Engines.
3. To study the heat transfer phenomena in IC engines.
4. To understand the thermal conductivity of the walls of the heat engines.
5. To learn practically the laws governing heat emission

Expt. No.	Experiments
1	Valve timing diagram
2	Port timing diagram
3	Actual p-v diagrams for IC engines
4	Performance test on 4 – stroke Diesel Engine
5	Heat Balance Test on 4 – stroke Diesel Engine.
6	Morse Test on Multi-cylinder Petrol Engine
7	Retardation Test on a Diesel Engine
8	Determination of Flash Point and Fire Point of various fuels / lubricants
9	Thermal conductivity measurement using guarded plate apparatus.
10	Thermal conductivity measurement of pipe insulation using lagged pipe apparatus.

11	Determination of heat transfer coefficient under natural convection from a vertical cylinder
12	Determination of Thermal conductivity of composite wall
13	Heat transfer from pin-fin apparatus (natural & forced convection modes)
14	Determination of Stefan – Boltzmann constant.
15	Determination of emissivity of a grey surface.
16	Effectiveness of Parallel / counter flow heat exchanger.

Course Assessment Method

(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Conduct tests on heat conduction apparatus and evaluate thermal conductivity of materials.	K3
CO2	Demonstrate tests on natural and forced convective heat transfer apparatus and evaluate heat transfer coefficient	K2
CO3	Conduct tests on radiative heat transfer apparatus and evaluate Stefan Boltzmann constant and emissivity.	K3
CO4	Conduct tests to evaluate the performance of parallel/counter flow heat exchanger apparatus	K3
CO5	Understand the laws governing heat emission	K1

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	-	-	-	-	-	2	3	2	-	2
CO2	2	1	-	-	-	-	-	2	3	2	-	2
CO3	2	2	-	-	-	-	-	1	3	2	-	3
CO4	1	1	-	-	-	-	-	2	3	2	-	2
CO5	2	2	-	-	-	-	-	2	3	2	-	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A course in thermal Engineering	Kothandaraman.C.P., Domkundwar. S,Domkundwar. A.V	Dhanpat Rai & sons	Fifth Edition 2016
2	Internal Combustion Engines	Ganesan V	Tata McGraw Hill, New Delhi	6 th Edition, 2012
3	Engineering Thermodynamics	Nag.P.K.	Tata McGraw Hill, New Delhi	6th Edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Engineering Thermodynamics: Fundamentals and Applications”	Natarajan, E	Anuragam Publications, Chennai	2 nd Edition, 2014
2	Thermodynamics - An Engineering Approach	Cengel, Y and M. Boles,	Tata McGraw Hill	,9th Edition, 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_me31/preview
2	https://onlinecourses.nptel.ac.in/noc22_me97/preview
3	https://archive.nptel.ac.in/courses/112/103/112103313/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.

- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

AGRICULTURE ENGINEERING

SEMESTER S6

**POST HARVEST ENGINEERING OF CEREALS, OIL SEEDS AND
HORTICULTURAL CROPS**

Course Code	PCAGT601	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Unit Operations in Agricultural Processing	Course Type	Theory

Course Objectives:

1. To provided fundamental knowledge in cereals, pulses and oilseeds processing
2. To understand the performance evaluation of different types of cleaners, separators, and size reduction machines
3. To learn various unit operations involved in processing of horticultural produce
4. To learn preservation, packaging, storage and quality evaluation of cereals, pulses oilseeds and horticultural products
5. To enhance knowledge in operation of post-harvest equipment

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	FUNDAMENTALS OF POST HARVESTING Production and utilization of cereals and pulses, grain structure of major cereals, pulses and oilseeds and their milling fractions; grain quality standards and physico-chemical methods for evaluation of quality of flours. Cleaning and grading, aspiration, scalping; Size reduction: principle, Bond's law, Kick's law, Rittinger's law, procedure (crushing, impact, cutting and shearing), Size reduction machinery: Jaw crusher, Hammer mill, Plate mill, Ball mill.	11

2	PROCESSING OF CEREALS AND PULSES Pre-milling treatments and their effects on milling quality; parboiling and drying, Drying: moisture content and water activity; Free, bound and equilibrium moisture content, isotherm, hysteresis effect, EMC determination, Psychrometric chart and its use in drying, Drying principles and theory, Different methods of drying, conventional, modern and integrated rice milling operations; wheat roller flour milling; processes for milling of corn, oats, barley, gram, pulses, paddy and flour milling equipments.	11
3	PROCESSING OF OILSEEDS Oilseed processing: mechanical expression, screw press, hydraulic press, solvent extraction methods, Extraction and refining of oil from coconut, soya bean, sunflower and rice bran. Packaging of processed products, design characteristics of milling equipments; selection, installation and their performance, BIS standards for various processed products	11
4	PROCESSING OF HORTICULTURAL CROPS Importance of processing of fruits and vegetables, spices, condiments and flowers. Characteristics and properties of horticultural crops important for processing. Peeling, Slicing, Blanching, Chilling, freezing and Packaging. Cold storage heat load calculations and cold storage design, refrigerated vehicle and cold chain system. Dryers for fruits and vegetables, Osmo-dehydration, General methods of preservation of fruits and vegetables, Flowcharts for preparation of different finished products, Important parameters and equipment used for different unit operations.	11

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Acquire fundamental knowledge in cereals, pulses and oilseeds processing	K2
CO2	Understand the performance evaluation of different types of cleaners, separators, and size reduction machines	K1
CO3	Understand various unit operations involved in processing of horticultural produce	K2
CO4	learn preservation, packaging, storage and quality evaluation of cereals, pulses oilseeds and horticultural products	K2
CO5	Have enhanced knowledge in operation of post-harvest equipment	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	1	1	2	-	-	-	2	1
CO2	2	2	3	1	1	1	2	-	-	1	-	1
CO3	2	2	3	1	1	1	-	1	-	-	2	1
CO4	2	2	3	1	1	1	-	-	2	-	2	1
CO5	2	2	3	1	1	1	-	2	-	-	2	1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Post harvest technology for Cereals, Pulses and oilseeds	Chakraverty, A.	Oxford & IBH publication Pvt Ltd, New Delhi	Third Edition, 2000
2	Unit operations of Agricultural Processing	Sahay, K.M., and Singh, K.K	Vikas publishing house Pvt. Ltd., New Delhi	1994
3	Unit operations in Food Processing	Earle, R.L.	Pergamon Press, Oxford, U.K	1985
4	Rice Processing and Allied Operations.	Dash, S.K., Bebartta, J.P. and Kar, A	Kalyani Publishers, New Delh	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Agriculture Processing	Pande, P.H	Kalyani Publishers, Ludhiana	1994
2	Agricultural Process Engineering	Henderson, S.M. and R.L. Perry	John Wiley and Sons, New York	1955
3	Handbook of Postharvest Technology, Cereals, Fruits, Vegetable, Tea and spices	Amalendu Chakraverty, Arun S Mujumdar, G S Vijaya Raghavan, Hoshali S Ramaswamy	Marcel Dekker, inc.	2003
4	Unit Operations in Food Engineering	Albert Ibarz and Gustavo V. Barbosa-Cánovas	CRC Press LLC, Florida	2003
5	Food Processing: Principles and Applications	<u>Hosahalli S. Ramaswamy, Michele Marcotte</u>	CRC Press	2005
6	Post Harvest Engineering of horticultural crops.	Sudheer, K P. and Indira, V.	New India Publishing House.	2007.
7	Drying and Storage of Cereal Grains	B K Bala	Wiley Blackwell	2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/126105023
2	https://nptel.ac.in/courses/126105023
3	https://nptel.ac.in/courses/126105023
4	https://nptel.ac.in/courses/126105023

SEMESTER S6
DAIRY AND FOOD ENGINEERING

Course Code	PCAGT602	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Food Processing and Preservation	Course Type	Theory

Course Objectives:

1. To expose the students to the fundamental knowledge of food, its properties and different methods of food processing
2. To understand the chemical, physical, and biological properties of milk and dairy products
3. To learn the methods and technologies used in the processing of milk and the production of various dairy products such as cheese, butter and ice cream.
4. To gain knowledge about the design, operation, and maintenance of equipment and machinery used in the dairy and food industry, including pasteurizers, sterilizers, homogenizers, separators, evaporators, filtration and dryers.
5. To equip the students to design and layout of dairy plant, sanitation and effluent treatment in dairy industry

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	PROPERTIES AND PROCESSING OF MILK Dairy Industry-importance and status; Dairy development in India, Milk Types – Composition and thermal and chemical properties of milk and milk products, Production of high-quality milk, Method of raw milk procurement and preservation - Processing	9
2	UNIT OPERATION IN MILK PROCESSING Process flow charts for product manufacture, Unit operation of various dairy	

	and food processing systems. Principles and equipment related to receiving of milk, pasteurization, sterilization, homogenization, centrifugation and cream separation.	9
3	PROCESSING AND PRESERVATION OF FOODS Evaporation: principles, types of evaporators, steam economy, multiple effect evaporation, multiple effect evaporation, vapour recompression. Principles of drying, spray drying, drum drying, freeze drying, Filtration: principle, types of filters; Membrane separation, RO, Nano-filtration, Ultra filtration and Macro-filtration, equipment and applications- Dairy plant design and layout-CIP, Dairy effluent treatment plant	9
4	PROCESSING OF DAIRY PRODUCTS Manufacture of Milk Powder, Processing of Milk Products - Condensed Milk - Skim milk - Butter milk - Flavoured Milk, Whey, casein, yoghurt and paneer, Manufacture of Butter: principles, butter churn; Preparation methods and equipment for manufacture of Cheese, Ghee, ice creams and frozen desserts - standards for milk and milk products - Packaging of Milk and Milk Products	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Gain fundamental knowledge of food, its properties and different methods of food processing	K2
CO2	Understand the chemical, physical, and biological properties of milk and dairy products	K1
CO3	Learn the methods and technologies used in the processing of milk and the production of various dairy products such as cheese, butter and ice cream.	K2
CO4	Gain knowledge about the design, operation, and maintenance of equipment and machinery used in the dairy and food industry, including pasteurizers, sterilizers, homogenizers, separators, evaporators, filtration and dryers.	K2
CO5	Learn design and layout of dairy plant, sanitation and effluent treatment in dairy industry	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	1	2	2	1	1	2	-	-	-	2	-
CO2	-	-	3	3	1	2	2	-	-	1	-	-
CO3	-	-	2	2	2	1	-	1	-	-	2	-
CO4	1	1	2	2	1	1	-	-	2	-	2	-
CO5	1	1	3	3	1	1	-	2	-	-	2	-

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Essentials of Food Process Engineering	Chandra Gopala Rao	B.S.Publications, Hyderabad	2006
2	Dairy Science and Technology	Walstra. P., Jan T. M. Wouters., Tom J. Geurts	CRC Press	2005
3	Technology and Engineering of Dairy Plant Operations	Ananthakrishnan, C.P., and Sinha, N.N	Laxmi Publications, New Delhi	1999
4	Outlines of Dairy Technology	Sukumar De	Oxford University Press	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Dairy Plant Engineering and Management	Tufail Ahmed	Kitab Mahal Publishers, Allahabad	1997
2	Introduction to Food Engineering	Singh, R P& Heldman, D. R	Academic Press	1993
3	Fundamentals of Food Process Engineering	Toledo, R.T	CBS Publishers and Distribution, New Delhi	1997
4	Food Processing and Preservation	Subbulakshmi.G., and Shobha A. Udipi	New Age International Publications, New Delhi	2007

Video Links (NPTEL, SWAYAM...)

Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc19_ag05/preview
2	https://onlinecourses.nptel.ac.in/noc19_ag05/preview
3	https://onlinecourses.nptel.ac.in/noc19_ag05/preview
4	https://onlinecourses.nptel.ac.in/noc19_ag05/preview

SEMESTER S6

RENEWABLE ENERGY AND GREEN TECHNOLOGY

Course Code	PEAGT631	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the importance and role of renewable sources of energy in the energy sector
2. To understand the production of energy through solar photovoltaic system
3. To learn about wind energy and turbines
4. To understand various aspects of bio energy and bio fuels
5. To understand the characteristics and basic principle of green technology

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to energy- Classification-Energy and environment- Concept of Renewable Energy Sources (RES)-Clean Development Mechanism-Role of renewable energy for mitigation of Global warming. Solar Energy: Fundamentals and basic principles- Solar radiation measurement, Basic Principles of Solar thermal energy conversion, Flat plate and Concentrating collectors, different solar thermal devices, Applications, and gadgets- Solar drying, Solar still	9
2	Solar Photo voltaic electricity production: Principles of Photo voltaic energy production-p-n junctions, Solar cells, PV Systems- Cell characteristics. Wind Energy: Energy available in wind, General formula, Lift and drag.	9

	Basics of Wind energy conversion, Power coefficient- Betz limit- Operational parameters of wind turbines-torque coefficient- tip speed ratio. Types of wind turbine rotors	
3	Bio-energy: Thermo-chemical energy conversion of biomass – Biomass combustion- Combustion of Biomass and stoves -Biomass gasification, Types of gasifiers. Biochemical energy conversion of biomass: Anaerobic digestion process-types of biogas plants- Basic design aspects of Biogas plants- operational and environmental parameters affecting biogas generation. Liquid bio-fuels, Basic principles for production of alcohol and biodiesel. Use of vegetable oils as fuel - Bio-diesel - general principles of the technology- production and use. Alcohol for fuel purpose- production processes and technology. Hydrogen as fuel – Hydrogen production by bio-photolysis – Biogas Fuel cells	9
4	Biomass combustion: Combustion, gasification and pyrolysis of biomass - Basic principles. Technology and equipment for biomass combustion- Chemistry of gasification- Different types of gasifiers– Thermal applications of producer gas- Use of producer gas in engines Concept of Green Technology – Importance – Green fuels – Zero emissions target – Carbon credit and trading – Green energy production – Fuels cells and Biological Photovoltaics – Eco-friendly materials – Green building – Principles aspects Certification systems – Green Rating for Integrated Habitat Assessment (GRIHA) etc.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the importance and role of renewable sources of energy in the energy sector	K1
CO2	Understand the production of energy through solar photovoltaic system	K2
CO3	Explain about wind energy and its turbines	K3
CO4	Understand various aspects of bio energy and bio fuels	K2
CO5	Understand the characteristics and basic principle of green technology	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2							1		
CO2	3	1	1			2				1		
CO3	3	1	2			2				1		
CO4	3	1	1			2				1		
CO5	3	1	2							1		

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Renewable energy resources	Tiwari and Ghosal	Narosa publication	2004
2	Non-conventional Energy Sources	G.D. Rai	Khanna Publication	6th Ed., 1988
3	Fundamentals of Renewable Energy	P.S. James and D.S. Khatawkar	Scientific Publishers	1st Ed., 2023

Reference Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Renewable Energy Resources	Twidell & Weir	Routledge	3rd Ed., 2015
2	Solar Energy	S. P. Sukhatme and J K. Nayak	Tata McGraw-Hill	4th Ed., 2017
3	Non-Conventional Energy Systems	K M. Mittal	A H Wheeler Publishing Co Ltd	1999
4	Renewable Energy Technologies	MK Ghosal	Narosa publication	Reprint 2018
5	Biomass, energy and environment: a developing country perspective from India	Ravindranath NH	Oxford University Press	1995
6	Energy Efficient Buildings in India	Mili Mujumdar	The Energy and Resources Institute, TERI	2009
7	Alternative Building Materials Technology	Jagadhish, Reddy and Rao	New Age International Private Limited	2017
8	Green Architecture: Advanced Technologies and Materials	Osman Attmann	McGraw-Hill	2010

SEMESTER S6

ENVIRONMENTAL IMPACT ASSESSMENT

Course Code	PEAGT632	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the students, the concept of EIA and its development and need.
2. To provide an awareness about administrative and legal measures for the protection of environment and environmental clearance procedures in India.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition, Need for EIA, Evolution of EIA: Global & Indian scenario - Environmental legislations in India- The Water (Prevention & Control of Pollution) Act 1974, The Air (Prevention & Control of Pollution) Act 1981, The Environmental (Protection) Act 1986- Environmental standards for water, air and noise quality- EIA Notification 1994 – EIA Revised notification 2006.	9
2	Environmental clearance process in India: Screening, Scoping, Public Consultation, Appraisal- Form1-Category of projects- Generic structure of EIA report- Terms of Reference (ToR) -Types of EIA: strategic, regional, sectoral, project level- Rapid EIA and Comprehensive EIA- Initial Environmental Examination (IEE).	9
3	EIA methodologies: Ad hoc, checklist, matrix, network and overlay- Impact Prediction, Evaluation and Mitigation-Prediction and assessment of the impact on water, air, and noise environment- assessment of ecological impacts and socio economic impacts. Impacts of modern agriculture and	9

	deforestation on environment.	
4	Environmental Management Plan (EMP): Goal and purpose- Importance of EMP- Content of an EMP- Role of environmental monitoring program. Environment Audit: need for audit- audit types and benefits- environmental audit procedure ISO 14001 standards: Importance, salient features - Stages in implementation- Benefits. EIA case studies (Indian) - a hydroelectric project, a agricultural production unit, food processing, fertilizer unit, irrigation project.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the need for minimizing the environmental impacts of developmental activities	K2
CO2	Outline environmental legislation & clearance procedure in the country	K1
CO3	Apply various methodologies for assessing the environmental impacts of any developmental activity	K3
CO4	Prepare an environmental impact assessment report	K2
CO5	Conduct an environmental audit	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	2	2	-	-	-	-	-
CO2	-	-	-	-	-	2	-	-	-	-	-	-
CO3	2	-	-	3	2	-	3	-	-	-	-	-
CO4	-	-	-	2	-	2	2	3	-	3	-	-
CO5	-	-	-	2	1	-	2	2	-	2	-	-

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental Impact Assessment	Larry W Canter	McGraw Hill Inc., New York	2011
2	Environmental Impact Assessment	Y. Anjaneyulu	B.S Publications	2003
3	Environmental Impact Assessment (Practical Solutions to Recurrent Problems)	Lawrence, David P	Wiley International, New Jersey	2001

SEMESTER S6

FOOD QUALITY & SAFETY

Course Code	PEAGT633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarize the students, the concept of EIA and its development and need.
2. To provide an awareness about administrative and legal measures for the protection of environment and environmental clearance procedures in India.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Food hygiene – introduction, definition, basic aspects of personal hygiene, disease transmission - food handling -preventing food borne illnesses, education of food handler in handling and serving food, importance of personal hygiene of the food handler, pest control - food institutions – sanitation of food institutions.	9
2	Food packaging – definition, functions, classification, laws related to packaging. Food labeling: standards, purpose, description types of labels, labeling regulation barcode, nutrition labeling – nutrition labeling, health claims, and mandatory labeling provision.	9
3	Food adulteration – definition of adulteration, adulterants, types of adulterants, food laws – essential commodities act, milk and milk products order, food standardisation and regulation agencies in India- agencies at state level, central food laboratories, HACCP, powers of food inspectors. Food spoilage -	9

	physical, chemical and microbiological spoilage of foods. Conduct a hazard analysis, CCP identification, establish critical limits for each CCP, establish CCP monitoring procedures, establish corrective actions procedures, establish procedures for HACCP verification and validation, documenting the HACCP Program.	
4	Introduction to food preservation- methods of preservation, general principles and applications of food preservation. Preservation by use of temperature - preservation by use of high and low temperature. Preservation by drying- preservation by drying and use of salt and sugar - enhancement of foods – food fortification, enrichment, supplementation, fermentation, germination, pre- and probiotics and organic foods.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Familiarize to apply protocol for safe food handling techniques, water and waste management	K2
CO2	Understand the role of food packaging and the importance of Nutrition labelling	K1
CO3	Analyse consequences of food poisoning and infection on the health of individuals	K3
CO4	Understand the basic principles food preservation methods	K2
CO5	Understand the procedure to implement HACCP program to any food industry	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2	1								2
CO2	3	2	2	1	1		1					
CO3	3	1	2	1	2		1					2
CO4	3	2	2	2	2		1					
CO5	3		1				1					2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Food Additives, Characteristics, Detection and Estimation	Mahindra N. S,	APH Publishing Corporation, New Delhi	2008
2	Perspectives in Nutrition	Ward law G.M, Hamp J S	7th edition, Mc Graw Hill	2007
3	The Food Safety and Standards Act along with Rules and Regulations	Anonymous	Commercial Law Publishers (India) Pvt Ltd., Delhi	2011

SEMESTER S6
DESIGN OF AGRICULTURAL IMPLEMENTS & MACHINES

Course Code	PBAGT604	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Farm Machinery and Equipment Tractor systems and control	Course Type	Theory

Course Objectives:

1. To understand the basic concept of machine design.
2. To understand the strength and stress failures of different joints.
3. To learn the design of shaft, keys, belts and gears.
4. To understand the design considerations of tillage implements.
5. To understand the design considerations of sowing, harvesting and threshing equipment / machines

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Understanding of design concepts, Phases of design, design considerations. Common engineering materials and their mechanical properties. Types of loads and stresses, Failure theories, factor of safety, selection of allowable stress. Stress concentration. Elementary fatigue and creep aspects. Cotter joints, knuckle joint and pinned joints, turnbuckle. Design of welded joints subjected to static loads.	9
2	Design of threaded fasteners subjected to direct static loads, bolted joints loaded in shear and bolted joints subjected to eccentric loading. Design of shafts under torsion and combined bending and torsion. Design of keys. Design of lever, Design of Jaw coupling, sleeve, and rigid flange couplings. Design of helical and leaf springs. Design of flat belt and V-belt drives and pulleys.	9

3	Design of gears- Spur gear, Design of screw motion mechanisms like screw jack, lead screw, etc. Selection of anti-friction bearings -Ball bearings Design considerations for tillage implements such as ploughs, harrows and cultivators. Design of mould board plough, analysis of forces acting on plough bottom by the cut soil slices, Disc implement types and design of disc for different disc tools, spacing of disc in multi-disc tools. Force analysis of vertical and inclined disc tools. Specific work of rotary tools and machine and dynamic forces. Reliability criteria in design and its application.	9
4	Design consideration of sowing and planting equipment, design of seed metering mechanism, design parameters of seed and fertilizer box and frame, design of seed metering mechanism of planters. Fertilizers handling equipment and their design parameters and requirements. Design consideration of threshing and harvesting equipment with reference to types of cylinder and safety measures. Design and selection of matching power unit. Safety devices and ergonomical considerations for the design and development of tractors & farm implements.	9

Suggestion on Project Topics

Course Assessment Method (CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 2 sub divisions. • Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic concept of machine design.	K1
CO2	Understand the strength and stress failures of different joints.	K2
CO3	Learn the design considerations of shaft, keys, belts and gears.	K2
CO4	To understand the design aspects of tillage implements.	K3
CO5	To understand the design considerations of sowing, harvesting and threshing equipments / machines.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	3	-	-	-	3	-	3	-	1	-
CO2	2	3	-	-	2	1	-	2	-	1	-	3
CO3	1	2	-	1	-	1	-	2	-	2	-	2
CO4	-	3	-	-	2	-	-	-	-	-	2	2
CO5	-	-	3	-	-	-	3	2	3	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Tractors and their Power Units	Barger EL, Liljedahl JB & McKibben EC	Wiley Eastern	
2	Agricultural Machines	Bernacki C, Haman J & Kanafajski CZ.	Oxford & IBH	
3	Theory, Construction and Calculations of Agricultural Machinery Vol. I.	Bosoi ES, Verniaev OV & Sultan-Shakh EG.	Oxonian Press	
4	A Text Book of Machine Design	Sharma PC & Aggarwal DK.	Katson Publishing House	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Pesticides Application Equipments	Bindra OS & Singh Harcharan	Oxford & IBH	
2	Agricultural Machines. Theory of Operations. Computing and Controlling Parameters and the Condition of Operation.	Klenin NI, Popov IF & Sakoon VA.	Amrind Publ.	
3	Tractor Implements System.	Ralph Alcock.	AVI Publ.	
4	Soil Dynamics in Tillage and Traction.	William. R Gill & Glen E Vanden Berg.	US Govt. Printing Office, Washington, D.C.	
5	Rotary Soil Working Machines Construction, Calculation and Design.	Yatsuk EP.	American Publ.Co.	
6	Pesticide Application Equipment for Use in Agriculture. Vol. II. Mechanically Powered Equipment.	Thornhill EW & Matthews GA.	FAO Rome	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/PEojc_K7u9U?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/yTrBEUvIqxI?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k
2	https://youtu.be/DsipD8fLjuU?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/jy-5oa47wrQ?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/7PtetaYPe5A?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/eHa1lTByJn4?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k
3	https://youtu.be/pO3EVgEZ-Os?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/FMl_s0Y1BaQ?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/0ry4B5LxUzw?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/2owpYC1hOiQ?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/2TSaTDrviIc?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/hzwENY96vEk?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/nQZLsZqwXLI?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/_xwFrWdEW_U?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k
4	https://youtu.be/XvBw-ZFoMKo?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/Q5PdBZ1DCYw?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/hXXo25FCuWQ?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/SqTsQzueV8I?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/6Ry2XKwCEFY?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/ig3NPqQcD6M?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k https://youtu.be/oUZ18-UZ7n0?list=PLbRMhDVUMngfpJp_tkeFAy_qF20vlwn3k

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S6

PLASTIC APPLICATION IN AGRICULTURE

Course Code	OEAGT611	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand about sustainability in agricultural processes, promoting improved yields and resource conservation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Types and quality of plastics used in soil and water conservation, production agriculture and post-harvest management. Quality control measures. Bore-well casing and subsurface drainage. Drip and sprinkler irrigation systems. Use of polymers in control of percolation losses in fields. Soil conditioning - soil solarisation, effects of different colour plastic mulching in surface covered cultivation. Nursery management - Use of plastics in nursery raising, nursery bags, trays etc.	9
2	Plastics as cladding material, green / poly / shade net houses, wind breaks, poly tunnels and crop covers. Plastic nets for crop protection - anti insect nets, bird protection nets. Plastic fencing. Plastics in drying, preservation, handling and storage of agricultural produce, innovative plastic packaging solutions for processed food products. Plastic cap covers for storage of food grains in open.	9
3	Use of plastics as alternate material for manufacturing farm equipment and machinery. Plastics for aquacultural engineering and animal husbandry -	9

	animal shelters, vermi-beds and inland fisheries. Silage film technique for fodder preservation. Agencies involved in the promotion of plasticulture in agriculture at national and state level. Human resource development in plasticulture applications.	
4	Water management - use of plastics in in-situ moisture conservation and rain water harvesting. Plastic film lining in canal, pond and reservoir.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the concept of plasticulture.	K2
CO2	Understand about the plastic pipes used for irrigation water management	K1
CO3	Describe controlled environmental cultivation.	K3
CO4	Explain about use of plastics as alternate material for manufacturing farm equipment and machinery.	K2
CO5	Explain the present status and future prospective of plasticulture in India.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2	1								2
CO2	3	3		1	1		1					
CO3	3	2	1	1								2
CO4	3	2	1	1	1		1					
CO5	3		1				1					2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advances in Protected Cultivation	Brahma Singh, Balraj Singh, Naved Sabir and Murtaza Hasan	New India Publishing Agency, New Delhi	2014
2	Plastics in Agriculture.	Dubois	Applied Science Publishers Limited, Essex, England	2007
3	Plastics Fundamentals, Properties, and Testing	Manas Chanda, Salil K. Roy	CRC Press	2008
4	Polymers in Agriculture and Horticulture. RAPRA Review Reports : Vol. 15, No. 2,	Brown, R.P.	RAPRA Technology Limited, U.K.	2004

SEMESTER S6

LANDSCAPE DESIGN AND MANGEMENT

Course Code	OEAGT612	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To equip students to create visually appealing, ecologically conscious outdoor spaces and fosters skills applicable to various fields such as architecture in agriculture, urban planning, and environmental science.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Standards, Bye laws and planning regulations applicable to building and landscape development. The role of statutory and regulatory bodies such as the Municipal Corporation, N.D.M.C, D.D.A and Urban Art commission etc. Implementation process Sequence of activities from inception to completion: agencies involved at each stage, their professional relationships, and obligations. Co-ordination of agencies and activities on site. Practical examples.	9
2	Contract Procedure; Criteria for selecting contractors: the process of calling tenders. Comparison of various kind of tenders with regard to objectives, utility, and appropriateness. Tender Documentation and evaluation of tender; negotiations with contractors. Contract Documentation: Forms of contract; General and special conditions, specifications, Bill of quantities; significant	9

	clauses pertaining to defects, maintenance, arbitrations, etc. Parties to the contract; their roles, contractual relationships, and legal obligations.	
3	Preparation of a maintenance plan: watering, fertilization, aeration, mulching, edging, pest control, pruning, and weed control. Design for irrigation, drainage, electrical etc, Methods and materials involved in maintenance of the landscape features. Maintenance bids and services involved.	9
4	Progress evaluation and monitoring: various kinds of estimates, review and updating, simple examples of pert charts and bar diagrams. Site documentation: importance of written records. Site instruction book, periodic reports, visual records, bar charts etc. Techniques of inspection and quality control; visits to site under development.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the Regulations and Legal Aspects Codes.	K2
CO2	Understand the concept of Construction administration.	K1
CO3	Describe about construction documents.	K3
CO4	Explain about landscape maintenance.	K2
CO5	Explain the maintenance plans	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2	1								2
CO2	3	2	2	1	1		1					
CO3	3	2	1	1						2		2
CO4	3	2	1	1	1		1					
CO5	3		1				1					2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Professional practice of landscape architecture, Van Nostrand Reinhold,	Walter Rogers	CRC Press	1997
2	Landscaping, Principles and Practices	Jack.E.Ingels	Delmar publishers inc	2000
3	Landscape handbook of Tropical Landscape	W.F.Hill	Garden Art Press	2008
4	Environmental Law	H.N.Tiwari	Allahad law agency	2004

SEMESTER: S6

UNIT OPERATIONS IN AGRICULTURAL PROCESSING

Course Code	PCAGL607	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To understand the various unit operations in cereals, pulses, fruits and oil seed processing industries
2. To gain practical experience and proficiency in using various machineries involved in post-harvest unit operations
3. To develop practical skills in operation and performance evaluation of fruit processing machineries
4. To develop technical skills on performance evaluation cleaning, separating and milling equipment
5. To gain sufficient knowledge on drying characteristics of grains and measurement of moisture content and EMC

Expt. No.	Experiments
1	Performance evaluation of different types of cleaners and separators
2	Performance evaluation of different size reduction machines
3	Determination of fineness modulus and uniformity index
4	Measurement of moisture content of grains: dry basis and wet basis

5	Drying characteristics of grains and drying constant
6	Determination of EMC (Static and dynamic method)
7	Rice mills and their performance evaluation
8	Pulse mills and their performance evaluation
9	Oil mills and their performance evaluation
10	Performance evaluation of peeler and slicer
11	Performance evaluation of juicer and pulper.
12	Performance evaluation of blanching equipment
13	Determination of firmness of fruit by penetrometer
14	Testing adequacy of blanching
15	Determination of fruit acids by titration and calculation of the sugar/acid ratio.
16	Moisture content determination of fruits and Vegetables, drying characteristics

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the various unit operations in cereals, pulses, fruits and oil seed processing industries	K1
CO2	Gain practical experience and proficiency in using various machineries involved in post-harvest unit operations	K2
CO3	Develop practical skills in operation and performance evaluation of fruit processing machineries	K3
CO4	Develop technical skills on performance evaluation cleaning, separating and milling equipment	K3
CO5	Gain sufficient knowledge on drying characteristics of grains and measurement of moisture content and EMC	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1		1	1						1
CO2	3	3	1	2	1	1						1
CO3	3	3	1	2	1	1						1
CO4	3	3	1		1	1						1
CO5	3	3	1		1	1						1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Post harvest technology for Cereals, Pulses and oilseeds	Chakraverty, A	Oxford & IBH publication Pvt Ltd, New Delhi	Third Edition, 2000
2	Unit operations of Agricultural Processing	Sahay, K.M., and Singh, K.K	Vikas publishing house Pvt. Ltd., New Delhi	
3	Unit operations in Food Processing	Earle, R.L	Pergamon Press, Oxford, U.K	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Food Process Engineering	Toledo, R.T	CBS Publishers and Distribution, New Delhi	
2	Handbook of Food Processing Equipment.	George D. Saravacos and Athanasios E. Kostaropoulos	Springer Science, Business Media, New York, USA.	2002
3	Unit Operations of Chemical Engineering,	Warren L. McCabe, Julian Smith, Peter Harriott	7th Ed. McGraw-Hill, Inc., NY, USA.	2004
4	Drying and Storage of Cereal Grains	B K Bala	Wiley Blackwell	2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://swayam.gov.in/nd1_noc19_ch32/...
2	https://youtube.com/playlist?list=PLbRMhDVUMngd9ZQul3t-OUxD4713C7QCh&si=iqXbDAJr840PL058
3	https://youtube.com/playlist?list=PLhWYf911na-XNr-totysuQubXQLXlt6BN&si=D-qOw8w2XUdLV3zr

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

AGRICULTURE ENGINEERING

SEMESTER S7

WATERSHED DEVELOPMENT AND MANAGEMENT

Course Code	PGAET741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. The student will get a comprehensive idea about watershed management.
2. The student will be capable of planning and execution of watershed management projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Watershed - introduction and characteristics. Watershed development - problems and prospects, investigation, topographical survey, soil characteristics, vegetative cover, land use practices and socio-economic factors. Watershed management - concept, objectives, factors affecting, watershed planning based on land capability classes, hydrologic data for watershed planning, watershed codification, delineation and prioritization of watersheds – sediment yield index. Water budgeting in a watershed.	9
2	Management measures - rainwater conservation technologies - in-situ and ex-situ storage, water harvesting and recycling. Dry farming techniques - inter-terrace and inter-bund land management. Integrated watershed management - concept, components, arable lands – agriculture and horticulture, non-arable lands - forestry, fishery and animal husbandry.	9

3	Effect of cropping systems, land management and cultural practices on watershed hydrology. Watershed programme - execution, follow-up practices, maintenance, monitoring and evaluation. Participatory watershed management - role of watershed associations, user groups and self-help groups. People's participation in watershed management.	9
4	Estimate preparation for watershed projects- Detailed estimate-Analysis of rates-Abstract of estimate. Planning and formulation of project proposal for watershed management programme including cost-benefit analysis. Preparation of detailed project report (DPR) for watershed projects.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the watershed characteristics and watershed development and its problems	K1
CO2	Explain watershed management concept, watershed planning, watershed codification, delineation and prioritization of watersheds, water budgeting in a watershed	K3
CO3	Understand management measures, dry farming techniques, integrated watershed management	K2
CO4	Study watershed programme, participatory watershed management, people's participation in watershed management.	K3
CO5	Preparation of watershed projects, planning and formulation of project proposal for watershed management programme including cost-benefit analysis.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2	1								2
CO2	3	2		1	1		1					
CO3	3	2	1	1					1			2
CO4	3	2	1	1	1		1					
CO5	3		1				1					2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Watershed Planning and Management	Rajvir Singh	Yash Publishing House, Bikaner	2016
2	Soil and Water Conservation Engineering	R. Suresh	Standard Publishers Distributors, Delhi-6	2020
3	Hydrology and Soil Conservation Engineering: Including Watershed Management	Ghanshyam Das	Prentice-Hall of India Learning Pvt. Ltd., New Delhi	2009
4	Hydrology: Principles, Analysis, Design	H. N. Raghunath	New Age International Publishers	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Watershed Management: Design and Practice	P. K. Singh	E-media Publications, Udaipur	2012
2	Watershed Management	V.V. Dhruva Narayana G. Sastry & U.S. Patnaik	ICAR, New Delhi	1997
3	Soil Conservation and Land Management	S. K. Datta	International Book Distributors, Dehradun	2006
4	Soil and Water Conservation and Watershed Management	S.C. Mahnot	International Books and Periodicals Supply Service. New Delhi	2012

SEMESTER S7

POST HARVEST ENGINEERING OF HORTICULTURAL CROPS

Course Code	PEAGT742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To acquaint and equip the students with processing and handling of fruits and vegetables and flowers
2. To minimize the post-harvest losses through value addition and post-harvest unit operations
3. To facilitate the design features of the equipment used for the processing of horticultural products
4. To understand various preservation, packaging and storage methods of horticultural produce
5. To expose the students to modern facilities in supply chain management of horticultural products

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Status-Importance, constituents of fruits and vegetables, Post harvest losses in fruits and vegetables, factors influencing losses, Enzymatic and textural changes during ripening.Engineering properties of horticultural crops. –. Post ha vegetables, spices, condiments and flowers, Characteristics and properties of horticultural crops important for processing, Peeling: Differentpeeling methods and devices (manual peeling, mechanical peeling, chemical peeling, and thermalpeeling), Slicing of horticultural crops: equipment for slicing, shredding, crushing, chopping,juice extraction, etc., Blanching: Importance and objectives; blanching methods, effects on food(nutrition, colour, pigment,	9

	texture).	
2	Chilling and freezing: Application of refrigeration in different perishable food products, Thermophilic, mesophilic & Psychrophilic micro-organisms, Chilling requirements of different fruits and vegetables, Freezing of food, freezing time calculations, slow and fast freezing, Equipment for chilling and freezing (mechanical & cryogenic), Effect on food during chilling and freezing, Cold storage heat load calculations and cold storage design, refrigerated vehicle and cold chain system, Dryers for fruits and vegetables: tray, drum, spray, pneumatic, fluidized bed dryer, and foam mat drying. Osmo-dehydration, freeze drying. Irradiation, surface coatings, chemical dips, preservatives.	9
3	Packaging of horticultural commodities, Packaging requirements (in terms of light transmittance, heat, moisture and gas proof, microorganisms. Different types of packaging materials commonly used for raw and processed fruits and vegetables products, bulk and retail packages and packaging machines, handling and transportation of fruits and vegetables, Packhouse technology, Minimal processing Ethylene evolution during storage, methods to prevent ethylene production, physiological loss of weight, methods of storage, Low temperature storage, evaporative cooled storage, controlled atmospheric storage, Modified atmospheric packaging. Pests, insects, rodents, fungi – their nature, occurrence and control. Modern techniques of infestation control. Application of quality control techniques.	9
4	Preservation Technology, General methods of preservation of fruits and vegetables, Brief description and advantages and disadvantages of different physical/ chemical and other methods of preservation, Flowcharts for preparation of different finished products, Important parameters and equipment used for different unit operations, Post harvest management and equipment for spices and flowers, Quality control in fruit and vegetable processing industry. Food supply chain.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the properties and post-harvest processing of horticultural crops	K1
CO2	Have knowledge on different post-harvest operations and processing methods of horticultural crops.	K2
CO3	Recognize the working principles of equipment and machineries involved in post-harvest unit operations of horticultural crops	K2
CO4	Gain sufficient knowledge on preservation and storage of horticultural crops	K3
CO5	Understand post-harvest handling and supply chain management of horticultural crops	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3					2				1
CO2	3	2	2									2
CO3	3	2	3					2				2
CO4	3	2	2									1
CO5	3	2	3					2				2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fruit Processing	Arthey, D. and Ashurst, P. R.	Chapman and Hall, New York.	1966
2	Postharvest physiology, handling and utilization of tropical and subtropical fruits and vegetables	Pantastico, E.C.B	AVI Pub. Co., New Delhi.	1975
3	Post-harvest Technology of Fruits and Vegetables,	Thomson, A. K	Blackwell Science Publication.	1996
4	Fruits and Vegetable Preservation – Principles and Practices	Srivastava, R.P. and Sanjeev Kumar	International book Distributing Co., Lucknow.	1998
5	Post Harvest Engineering of horticultural crops.	Sudheer, K P. and Indira, V	New india Publishing House.	2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Preservation of Fruits and Vegetables,	Girdhari Lal, G.S. Siddappa & G.L. Tandon (	ICAR, New Delhi.	1995
2	Post Harvest Technology of Horticultural Crops	Kader,A.A, R.F. Kasmire, M.S. Reid, N.E. Sommer & J.E. Thompson	Cooperative Extension University of California, Division of Agriculture and Natural resources.	1985
3	Processing of Fruits: Science & Technology (Vol.II) – Major Processed Products,	Somogyi,L.P, D.M. Barrett, Y.H. Hui	Technomic Publishing Co. Inc., Lancaster, USA.	1996
4	Commercial Vegetable Processing,	Luh,B.S and Woodroof J.G	AVI Publication, New York.	1988
5	Processing of Fruits: Science and Technology (Vol.I) – Biology, Principles and Applications,	Somogyi,L.P, H.S. Ramaswamy & Y.H. Hui	Technomic Publishing Co. Inc., Lancaster, USA.	1996

SEMESTER S7

MECHANICS OF TILLAGE AND TRACTION

Course Code	PEAGT743	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart basic knowledge in mechanics of tillage and traction by understanding the fundamental concepts.
2. To develop analytical and practical skills in farm machinery and implements.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to mechanics of tillage tools, engineering properties of soil, principles and concepts, stress strain relationship.	9
2	Design of tillage tools principles of soil cutting, design equation, force analysis, assessment of the dynamic properties of soils.	9
3	Application of dimensional analysis in soil dynamics and traction prediction equation, Methods of dimensional analysis, Development of prediction equations.	9
4	Introduction to traction and mechanics, off road traction and mobility, traction model, traction improvement, tyre size, tyre lug geometry and their effects, tyre testing. Soil compaction and plant growth, variability and application of GIS in soil dynamics.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Measure and utilize physical and mechanical properties of soil in order to interpret and predict soil stress-strain behaviour.	K2
CO2	Design and implement safe and cost-effective mechanical soil tillage systems for producing desired physical states	K3
CO3	Design and implement and cost-effective mechanical traction systems which produce specified performance and acceptable alteration of affected soil profiles	K3
CO4	Understand various soil failure criterion and their interpretation.	K1
CO5	Understand the need to learn and apply improved methodologies through continuing education.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2							1		
CO2	3	1	1			2				1		
CO3	3	1	2			2				1		
CO4	3	1	1			2				1		
CO5	3	1	2							1		

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books

Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Agricultural engineering soil mechanics	Mc Kyes E.	Elsevier	2012
2	Basic Soil mechanics	R. Whitlow	Pearson	2000
3	Soil Dynamics in Tillage and Traction.	Gill & Vandenberg.	Supdt. Of Documents, U.S. Govt. Printing Office, Washington, D.C.	1968
4	Fundamentals of Soil Physics	Daniel Hill.	Academic Press	1962
5	Agricultural soil mechanics	A. J. Koolen and H. Kuipers	Springer Science & Business Media	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Design of Soil Tilling Machines	Sineokov G. N.	INSDOC, New Delhi.	1977
2	Soil Mechanics in Engineering Practices	Terzaghi K & Peck Ralph B.	John Wiley & Sons	1967
3	Soil Cutting and Tillage	Mc Kyes E.	Elsevier	1985
4	Soil Mechanics	William Lambe T. and Whiteman Robert V.	John Wiley & Sons	1969

SEMESTER S7

PRECISION FARMING TECHNIQUES FOR PROTECTED CULTIVATION

Course Code	PEAGT751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the fundamentals techniques of precision farming principles
2. To understand the application of precision farming equipment in crop production

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Precision Farming — Concept and Applications - Role of electronics in agricultural engineering for precision agriculture. Basics of precision agriculture, tools for implementation of precision agriculture. Introduction of GIS/GPS positioning system for precision farming. Use of GIS and GPS in farm machinery and equipment. Precision farming concepts - Map based system - Real time system – Combination Map and real time system - components of PF – Site specific management - Constraints of PF.	9
2	Greenhouse—Concept, Classification and Uses - Greenhouse—Design and Construction - Greenhouse Heating System - Greenhouse of Micro-climate - Greenhouse—Growing Media - Polytunnels, Shade Nets, Plastic Nets and Plastic Mulching - Greenhouse Irrigation Methods	9
3	Site-specific nutrient management- weeds management - Agro-chemicals and fertilizer management, data sources and decision making for site-specific	9

	management. Grain quality and yield. Yield monitoring and mapping, soil sampling and analysis.	
4	Design of Drip and Micro-Sprinkler Irrigation Systems - Fertigation of Greenhouse Crops - Water Quality—Greenhouse Use - Greenhouse—Insect Pests and Disease Management	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the role of precision farming in agriculture	K1
CO2	Impart the principles and applications of protected cultivation	K2
CO3	Understand the precision farming concepts	K2
CO4	Understand about site specific management system	K2
CO5	Explain the application of micro-irrigation systems and greenhouse management system	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	1	3					1	2
CO2	2		2		3	3					2	2
CO3	2			2	3	3						2
CO4	1	1	1	1	2	3						2
CO5	1	1	1		2	3						2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Precision Farming Techniques for Protected Cultivation	SURESH, R. and NIRALA, S. K.	PHI India	First Edition, 2024
2	Textbook of Field Crops Production: Commercial Crops Vol. II	Dr. Rajendra Prasad	Indian Council of Agricultural Research, New Delhi	First Edition, 2006
3	Precision Agriculture	Brase, T.A.	Thomson Delmar Learning, New York.	First Edition, 2006
4	Precision in Crop Farming, Site Specific Concepts and Sensing Methods: Applications and Results.	Hermann, J.H.	Springer, Netherlands	First Edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Precision Agriculture Technology for Crop Farming.	Zhang, Q.	CRC Press, New York.	2015
2	Engineering Principles of Agricultural Machines.	Srivastava,A K., Carroll E.G., Roger P. R. and Dennis R.B.	American Society of Agricultural and Biological Engineers, USA.	2006
3	Irrigation: Theory and Practice.	Michael, A.M	Vikash Publishing House Pvt. Ltd., New Delhi.	2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/3JobwSAvEcw ; https://youtu.be/PKNWecbMVWk
2	https://youtu.be/4PTywZ_VWYA ; https://youtu.be/PKNWecbMVWk
3	https://youtu.be/N5V_3WdDvlM
4	https://youtu.be/r_lhap2hDdk ; https://youtu.be/_CyFdcokjMA

SEMESTER S7

UNIT OPERATIONS IN POST HARVEST ENGINEERING

Course Code	PEAGT752	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To acquaint and equip the students with different unit operations of food industries and their design features.
2. To develop skills to process and handle fruits and vegetables, minimizing post-harvest losses and understanding equipment design features for processing.
3. To equip students with post-harvest technology for major cereals, pulses, and oilseeds, emphasizing equipment

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Size reduction: Benefits, classification, determination and designation of the fineness of ground material, sieve/screen analysis, fineness modulus and uniformity index, principle and mechanisms of comminution of food, Rittinger's, Kick's and Bond's equations, work index, energy utilization; Size reduction equipment: Principle, types, crushers (jaw crushers, gyratory, smooth roll), hammer mills and impactors, attrition mills, burr mill, tumbling mills, ultra fine grinders, fluid jet pulverizer, colloid mill, cutting machines (slicing, dicing, shredding, pulping).	9
2	Drying: principle, moisture and its removal, utilities of drying, Moisture content and its measurements, Moisture content representation, Determination methods:	

	direct and indirect methods, Equilibrium moisture content (EMC), Method for determination of EMC: dynamic methods and static methods, Drying theory, methods of drying, drying rate periods, Mechanical drying types; thin layer drying and deep bed drying	9
3	Rice milling, parboiling, methods:traditional and modern methods, milling equipments, hullers, sheller, paddy separator, removal of bran, polisher, glazing, grading , milling efficiency, Pulse milling, unit operations, commercial scale methods (wet milling and dry milling, Jabalpur method, CFTRI method, Pantnagar process, CIAE method), milling efficiency, Milling of oilseeds: mechanical expression, screw press, hydraulic press, solvent extraction methods, preconditioning of oilseeds, refining of oil	9
4	Importance of processing of fruits and vegetables. Peeling: Different peeling methods and devices (manual peeling, mechanical peeling, chemical peeling, and thermal peeling), Slicing of horticultural crops: equipment for slicing, shredding, crushing, chopping, juice extraction, etc., Blanching: Importance and objectives; blanching methods, effects on food (nutrition, colour, pigment, texture), Physiological changes during ripening, maturity index, firmness, sugar acid ratio, respiration, pre-cooling	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Familiarization of equipment used in different unit operations	K1
CO2	Conceptualise moisture content, drying, working of different driers	K3
CO3	Understand the concepts of processing of cereals, pulses and oilseeds along with byproduct utilization	K2
CO4	Gain concept of size reduction principle and related equipment	K2
CO5	Understand the process and handle fruits and vegetables, minimizing post-harvest losses	K1

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2					1			2	
CO2	3	3	2					1			2	
CO3	3	3	2					1			2	
CO4	3	3	2					1			2	
CO5	3	3	2					1			2	

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Unit Operations of Agricultural Processing	K.M Sahey and K.K Singh	Vikas Publishing House pvt ltd	Second revised edition 2001
2	Fruits &Vegetables Preservation, Principles and Practices	R.P Srivastava Sanjeev Kumar	International book distributing co.	First edition,1994
3	Post Harvest Technology of Horticultural Crops	K.P Sudheer and Indira	New India Publishing agency	
4	Post Harvest Technology of Cereals, Pulses and Oil Seed	Chakraverty	CBS publishers and distributers pvt ltd	3 rd edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Food Engineering	D.G Rao	PHI learning private ltd	First edition
2	Food Engineering Principle and Practices	Sanjaya K Dash Pitam Chandra AbhijitKar	CRC press Taylor & Francis Group	First edition
3	Food Processing Principle and Applications	HosahalliRamaswamy Michele Marcotte	CRC press Taylor & Francis Group	First edition
4	An introduction to the Principles of Food Process Engineering	Richard L Earle	British Library cataloguing in publication data	Second edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/126/105/126105011/
2	https://archive.nptel.ac.in/courses/126/105/126105011/
4	https://archive.nptel.ac.in/courses/126/105/126105023/

SEMESTER S7

WASTE AND BYPRODUCTS UTILIZATION

Course Code	PGAET753	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To acquaint and equip the students with the techniques of utilization of agricultural waste and by-products and also about development of value-added products from wastes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Conversion processes: Thermo-chemical conversions, densification, combustion and gasification, extraction, biological conversions, anaerobic digestion, biochemical digestion process, digestion systems, energy from anaerobic digestion, cellulose degradation, fermentation process. Agricultural wastes as paper, boards and fuel.	9
2	Briquetting: Briquetted fuel from husk, hull and other wastes selection, design of briquetting machines. Utilization of shell, stem and stalk: Production of activated carbon. By-products of agro-industries: Rice mill, oil mill, cattle feed mill, valuable constituents and composition. Utilization of rice husk: Production of silica and cement from rice husk. Stabilization and storage of rice bran, extraction of rice bran oil.	9
3	By-products of oil refining: Fatty acids/soap stock, wax and gum, characteristics and utilization. Rice germ and broken rice. Production of starch and infant food, industrial uses of starch. By-products of oil milling: Oil cake and defatted oil	9

	cake, cattle feed and industrial uses. Utilization of starch and other industrial wastes: Microcrystalline cellulose, production of ethanol, wastes of tapioca starch industries, Tapioca thippi-utilization as fuel, extraction of starch by hydrolysis, utilization of starch for food, adhesives and feed purposes.	
4	By-products of sugar industry: Sugarcane tops, bagasse, molasses and press-mud, utilization as animal feed. By-products of fruits and vegetables-based agro-industries: Mango seed kernel and pineapple waste.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals and basics to reduce waste creation.	K1
CO2	Analyse types of operation in wastewater treatment.	K3
CO3	Create alternative approach to reduce agricultural waste.	K3
CO4	Understanding Process of composite materials from agricultural waste.	K2
CO5	Investigations on working of anaerobic digestion	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2							1		
CO2	3	1	1			2				1		
CO3	3	1	2			2				1		
CO4	3	1	1			2				1		
CO5	3	1	2							1		

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	ASAE Standards. Manure Production and Characteristics.	ASAE	ASAE	1984
2	Rice: Production and Utilization.	Bor SL. (Ed.).	AVI Publ.	1980
3	Food, Feed and Fuel from Biomass.	Chahal DS.	Oxford & IBH.	1991
4	Biotechnology and other Alternative Technologies for Utilisation of Biomass/Agricultural Wastes.	Chakraverty A.	Oxford & IBH	1989
5	Fuels from Biomass and Wastes.	Donald LK and Emert HG.	Ann. Arbor. Science Publ.	1981
6	Biomass Briquetting and Utilization	Srivastava PK, Maheswari RC and Ohja TP.	Jain Bros.	1995
7	Agricultural Waste Management Field Handbook.	USDA.	USDA.	1992

SEMESTER S7

DESIGN AND MAINTENANCE OF GREEN HOUSE

Course Code	OEAGT721	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To gain a comprehensive understanding of the principals involved in creating controlled environments for plant growth, fostering skills in architectural planning, climate control, and sustainable cultivation practices.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	History and types of greenhouses; importance, function, and features of green house; scope and development of greenhouse technology. Location, Planning, and various component of greenhouse; design criteria and calculation; constructional material and methods of construction; covering materials and its characteristics, solar heat transfer, solar fraction for green house, steady state analysis of green house.	9
2	Greenhouse heating, cooling, shedding and ventilation systems; Carbon Dioxide generation and monitoring and lighting systems, instrumentation & computerized environmental Control Systems.	9
3	Watering, fertilization, root substrate and its pasteurization, containers and benches, plant nutrition. Alternative cropping systems; plant tissue culture, chemical growth regulation; disease control; integrated pest management; postproduction quality and handling.	9

4	Cost analysis of greenhouse production; Applications of green house & its repair & maintenance	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination- 1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the types of green house.	K1
CO2	Understand construction and planning of green house.	K1
CO3	Describe about ventilation and environment control systems	K3
CO4	Explain about alternate cropping system and pest control	K2
CO5	Explain cost analysis, repair and maintenance of green house.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	P012
C01	3		2	1								2
C02	3	2	2	1	1		1					
C03	3	2	1	1			1					2
C04	3	2	1	1	1		1					
C05	3		1				1					2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No.	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1.	Greenhouse Technology	Tiwari G.N. and Goyal R.K	Narosa Publishing House	2011
2.	Greenhouse Advanced Technology	Hanan	CRC Press Inc	2012
3.	Greenhouse Operation & Management	Nelson P.V.	Wiley	2014
4.	Handbook of Greenhouse Technology	Radhamanohar	Narosa Publishing House	2000

SEMESTER S7

AGRIBUSINESS MANAGEMENT

Course Code	OEAGT722	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To enable the students to typically revolve around optimizing agricultural and related business operations to achieve sustainable growth, profitability, and social responsibility.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Agri-business - scope, characteristics, types. Management - importance, definition, management and administration, management thoughts, Small business - characteristics and stages of growth. Principles, forms of agri-business organizations, staffing, directing, supervision and motivation. Controlling - types, performance evaluation and control techniques. Management approaches - Profit Centered Approach, Management by objectives and Quality Circles. Strength, Weakness, Opportunities and Threat (SWOT) Analysis.	9
2	Functional areas of Agri-business - Production and Operations management - functions, planning physical facilities and managing quality. Agro-inputs and products inventory management – raw material procurement, inventory types, and costs. Marketing management- Marketing environment, marketing mix - Agricultural input marketing firms.	9
3	Forms of agri-business organizations - Role of lead bank in	9

	agribusiness finance - Financial management. Acquiring capital- Budget analysis. Concepts and determinants- Business project scheduling of raw material procurement - production management - launching products (branding, placement) - Input marketing promotion activities.	
4	Agricultural products - marketing promotion activities - product pricing methods. District Industries Centre - Consumer survey - Agricultural inputs retailing - Market potential assessment - types of distribution channels - Return on Investment - Personnel management. Recruitment, selection and training - Technology in Agri Business.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concepts of agricultural business and management.	K1
CO2	Explain the principles of business organization and management approaches.	K2
CO3	Describe Functional areas of Agri-business and products inventory management.	K3
CO4	Demonstrate Financial management and Budget analysis of agri business organizations	K2
CO5	Explain the product pricing methods and types of distribution channels	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	P01	P02	P03	P04	P05	P06	P07	P08	P09	P010	P011	P012
C01	3		2	1								2
C02	3	2		1	1		1					
C03	3	2	1	1					1			2
C04	3	2	1	1	1		1					
C05	3		1				1					2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Textbooks				
Sl. No.	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1.	Agri Business Management – Problems and prospects	Himanshu	Ritu Publications, Jaipur	2005
2.	Indian Agriculture and Agribusiness Management	Smita Diwase	Krishi resource Management Network, Pune	2004
3.	Projects: Preparation, Appraisal, Budgeting and Implementation	Chandra Prasanna	Tata McGraw Hill Publications, New Delhi	2001
4.	Marketing Management. Analysis, Planning and Control	Kotler, P	Prentice Hall Inc., New York	2001

SEMESTER 8

AGRICULTURE ENGINEERING

SEMESTER S8
SENSORS, AI AND ROBOTICS IN PRECISION AGRICULTURE

Course Code	PEAGT861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart basic knowledge on the modern technologies that are used for precision farming and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to sensors, sensors in precision farming; Spectral based sensors; Development of sensors and controls; Image analysis technology; Crop yield monitors; Yield Monitors: Components, Differential GPS Receiver, GNSS Receiver, mass flow sensors. Impact plates, measuring volume with a photoelectric sensor, volumetric flow sensing and alternatives. Grain moisture sensor, fan speed sensor, elevator speed sensor, header position, yield monitor data, cotton yield monitors	9
2	Introduction Remote sensing for precision agriculture and applications for crop monitoring, Developments in livestock and building control: Radio telemetry systems to remotely monitor and record physiological parameters.	9
3	Global navigation satellite system (GNSS). Satellite ranging: Accuracy, standards, components of GIS, data layers, map component, attribute table	9

	component, Function of a GIS, resolution. Data formats: Vector or raster. GIS for precision farming, data analysis, field calculator, convert to grid, interpolation, reclassification, image classification, band math, interpretation of analysis, farm management information systems, and crop intelligence	
4	Introduction to robotics in agriculture, Concept of robotics and classification and components, Applications of robotics in automation of farm operations, sensors and actuators, control electronics, Algorithms and programming for microcontroller and microprocessor-based robots, telemetry, remote control and autonomous operation of agricultural robots.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the main driving forces for automation in agriculture;	K1
CO2	Understand the basic principles and technologies of precision and site-specific farming.	K2
CO3	Understand the most commonly used sensors in precision agriculture and agricultural robotics and describe their working principles	K2
CO4	Selection of an appropriate sensor for a given application	K3
CO5	Visualize and analyze agricultural data	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2							1		1
CO2	3	1	1			2				1		1
CO3	3	1	2			2				1		1
CO4	3	1	1			2				1		
CO5	3	1	2							1		

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advances in Agricultural Machinery and Technologies.	Chen G. (ed).	CRC Press	2018
2	Sensors in Agriculture	Dimitrios Moshou (Editor)	MDPI	2019
3	Sensing Approaches for Precision Agriculture	Ruth Kerry, Alexandre Escolà	Springer	2021
4	Sensors Application In Agriculture	Dimitrios S. Paraforos, Spyros Fountas, Francesco Marinello	MDPI	2020
5	Robotics For Engineers- Concepts and Techniques	Kailash Chandra Mahajan, Prashant Patnaik, R. Kumar	Vikas Publishing House	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robotics in Agriculture	Katal Rajesh Singh, Anita Gehlot, Mahendra Swain, Nitish	New India Publishing Agency	2024
2	Smart Agriculture Modelling through Image Processing and Remote Sensing Technology	Hari Kumar Singh	Walnut Publications	2021
3	Engineering Principles of Agricultural Machines	Srivastava A K. 2006.	American Society of Agricultural and Biological Engineers	1993
4	Harvesting intelligence: transforming agriculture through artificial intelligence	AK Vala	Clever Fox Publishing	2023
5	Microprocessor-based agri instrumentation	Krishna Kant	PHI Learning	
6	Text Book Of Remote Sensing and Geographical Information Systems	M Anji Reddy	Bs Publications	3 rd Ed.

SEMESTER S8
DEVELOPMENT OF PROCESSED FOOD PRODUCTS

Course Code	PEAGT862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the transformation of raw materials into consumables.
2. To learn the fundamentals of innovative techniques, technological advancements and culinary expertise in modern food industries.
3. To provide the knowledge and skills needed to meet the dynamic demands of the food industry

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Applications of unit operations to the food industry, analytical processing concepts with regards to mass and energy balances, equipment involved in the commercially important food processing methods and unit operations; value addition to cereals like rice and wheat.	9
2	Parboiling of rice, quality of processed products of rice & wheat. Processing of pulses, spices, and condiments; extruded food product, fermented food product, frozen and dried product, technology of meat, fish and poultry products, technology of milk and milk products.	9

3	Technology of oilseeds and fat products, snack foods, Fruits, and vegetables product: candy, nutraceuticals, food product development trends, food additives and labeling.	9
4	Process equipment for thermal processing evaporation, dehydration, drying, blanching, pasteurization, distillation. Mechanical separation filtration, sieving, centrifugation, sedimentation; mechanical handling-conveying and elevation; size reduction and classification-mixing; kneading, blending.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the unit operations take place in food industries.	K1
CO2	Understand the concept of parboiling of rice and its quality in processing.	K2
CO3	Describe the technology used in oil seed and fat products.	K2
CO4	Explain about the equipment's used for thermal processing of food products	K3
CO5	Explain the mechanical separation and mechanical handlings of food products.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2	1								2
CO2	3	2	1	1	1		1					1
CO3	3	2	1	1					1			2
CO4	3	2	1	1	1		1					1
CO5	3		1				1					2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Unit operations of Agricultural Processing	Sahay, K. M. & K.K. Singh	Vikas Publishing House Pvt Ltd	First Edition, 2021
2	Post harvest technology of cereals, pulses, and oilseeds	Chakraverty, A	OXFORD & IBH PUBLISHING	Third Edition, 2019
3	Preservation of Fruits and Vegetables	Girdhari Lal, G.S. Siddappa & G.L. Tandon	ICAR, New Delhi	First Edition, 2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Processing Equipments for Agrl. Products	Carl.W. Hall	McGraw Hill Pub.Co	2015
2	Mechanism of action of Food Preservation Procedures	Gould, G	Elsivier applied Science	2006
3	Technology of Cereals	Kent, N.L	Oxford Pergamom	2007

SEMESTER S8

FOOD PACKAGING TECHNOLOGY

Course Code	PEAGT863	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand about packaging methods and the properties of packaging materials.
2. To understand the different packaging materials used in food industry, packaging machineries, modern packaging techniques etc.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction of storage: Factors affecting shelf life of food material during storage, Spoilage mechanism during storage, Definition, Requirement, Importance and scope of packaging of foods.	9
2	Introduction of packaging: Types and classification of packaging system, Advantage of modern packaging system, Different types of packaging materials used. Types of packaging: Different forms of packaging, metal container, Glass container, Plastic container, Flexible films, Shrink packaging, Vacuum and gas packaging. Modified and Controlled atmospheric packaging (MAP and CAP), Active and Intelligent Packaging, Aseptic Packaging.	9

3	Selection of packaging material: Packaging requirement and their selection for the raw and processed foods, Advantages and disadvantages of these packaging materials, Effect of these materials on packed commodities, Package testing, Printing, Labelling and lamination. Flexible Packaging: Moulding – Injection, blow, extrusion methods Package Testing: Functional and Mechanical tests Packaging and filling systems.	9
4	Economics of packaging: Economics of packaging, Performance evaluation of different methods of packaging food products, Their merits and demerits, Scope for improvements, Disposal and recycle of packaging waste.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the importance of storage of food materials.	K1
CO2	Understand the methods of packaging of perishable products.	K2
CO3	Impart knowledge of the types of packaging and their effect of quality maintenance.	K3
CO4	Understand the economics of packaging of food materials.	K3
CO5	Explain suitable packaging materials for various food products.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	3		1	1					1	2
CO2	2			1							1	2
CO3	2	2	1	2								2
CO4		1	2		2							2
CO5	1	1	1			1						2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Food Packaging	G. L. Roberston	AVI Publishing Co	Third Edition, 2000
2	Food Packaging	F. A. Paine and J. Y. Paine	Springer	First Edition, 1993
3	Fish Packaging Technology Materials and Methods	K. Gopakumar	Concept Publication, New Delhi	First Edition, 1998

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Food Packaging	S. Sachrow and R. C. Graffin	AVI Publishing Co	First Edition, 1998
2	Food Packaging Material	N. T. Crosby	Applied Science Publishers	Second Edition, 2020

SEMESTER S8 MECHATRONICS

Course Code	PEAGT864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the cutting-edge field of Mechatronics with a specialized focus on its applications in agriculture.
2. To learn the integration of mechanical systems, electronics, computer science, and control engineering is revolutionizing farming practices.
3. To understand how mechatronics is enhancing efficiency, sustainability, and precision in modern agriculture, paving the way for innovative solutions to address the challenges of the agricultural industry.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Mechatronics and Sensors in Agriculture-Introduction to Mechatronics-Basic Electronics and Electrical Components- Sensors and Actuators in Agricultural Systems-Signal Conditioning and Processing-Introduction to Microcontrollers and Microprocessors-Applications of Mechatronics in Agriculture. Control Systems in Agricultural Machinery-Introduction to Control Systems- Feedback Control Systems-PID Controllers	9

	in Agricultural Machinery-PLC (Programmable Logic Controller) Applications	
2	Modeling and Simulation of Agricultural Systems-Precision Farming and Automated Control. Robotics in Agriculture-Introduction to Agricultural Robotics- Types of Agricultural Robots- Robotic Vision Systems in Agriculture- Robot Manipulators and Grippers- Autonomous Navigation in Agricultural Robotics-Precision Agriculture using Robotics.	9
3	Data Analytics and Decision Support in Precision Agriculture- Introduction to Precision Agriculture- Data Acquisition and Management in Agriculture-Data Analytics Techniques for Crop Monitoring- GIS (Geographic Information System) in Agriculture-Decision Support Systems in Precision Agriculture-Case Studies in Precision Agriculture.	9
4	Emerging Trends in Mechatronics for Agriculture-Internet of Things (IoT) in Agriculture-Artificial Intelligence in Agricultural Systems- Drones and Unmanned Aerial Vehicles (UAVs) in Agriculture- Sustainable Agriculture Technologies- Future Prospects and Challenges in Agricultural Mechatronics.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of mechatronics and its relevance in agriculture.	K1
CO2	Identify and analyse various sensors and control systems used in agriculture machinery.	K2
CO3	Understand the basics of agricultural robotics and its applications.	K2
CO4	Acquire and manage data relevant to agriculture using modern technologies.	K3
CO5	Understand the applications of drones and UAVs in precision agriculture.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2	1								1
CO2	3	2	1	2	1		2					
CO3	2	2	2	1	2		1					2
CO4	3	2	2	2	1		2					
CO5	3		1				1					1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Mechatronics and Measurement Systems	David G. Alciatore and Michael B. Histanal.	Tata McGraw	Fourth Edition, 2021
2	Precision Agriculture Technology for Crop Farming	Qin Zhang and Huazhong Ren	CRC Press, New York	First Edition, 2021
3	Control Systems Engineering	Norman S. Nise	Wiley	First Edition, 2018
4	Programmable Logic Controllers	W. Bolton	ELSEVIER	Fourth Edition, 2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Geographic Information Systems and Science	Paul A. Longley	John Wiley & Sons	2005
2	Artificial Intelligence in Agriculture: Theories, Applications, and Future Directions	Valentina E. Balas	Wiley	2021

SEMESTER S8

HUMAN ENGG AND SAFETY

Course Code	OEAGT831	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart the fundamental knowledge to the student on the importance of human engineering and safety in the field of agriculture machinery.
2. To gain a general understanding of the safety features and regulation acts in farm machinery.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Human factors in system development, Concept of systems; basic processes in system development, performance reliability. Human performance. Information input process, visual displays, major types and use of displays, auditory and factual displays. Speech communications.	9
2	Biomechanics of motion, types of movements, Range of movements, Strength and endurance, speed and accuracy, human control of systems. Human motor activities, controls, tools and related devices.	9

3	Anthropometry: arrangement and utilization of work space, atmospheric conditions, heat exchange process and performance, air pollution.	9
4	Dangerous machine (Regulation) act, Rehabilitation and compensation to accident victims, Safety gadgets for spraying, threshing, Chaff cutting and tractor & trailer operation etc.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the importance of human factors and their application in system development.	K1
CO2	Know the effect of visual, auditory and factual displays in human performance.	K2
CO3	Understand the importance of optimum work-rest cycles in endurance.	K3
CO4	Ideally design the work space in accordance to anthropometry.	K2
CO5	Have the general understanding safety features and regulation acts in farm machinery.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2	1			1					2
CO2	3	3	2	1	1		1					1
CO3	3	1	2	1	2		1					2
CO4	3	2	2	2	2		1					2
CO5	3		1				1					2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human factors in engineering and design	Ernest and Mc Cormick, E.L.	Mc Graw Hill Co., New York.	2005
2	Fitting the task to the man	Grandjean, E	Taylor and Francis, London	2004
3	Tractors and power units	Liljedhal, J.B, Carleton, W.M, Smith, P.K and David, M	John Wiley and sons, New York	2001
4	Ergonomics, Man in his working environment	Murrel, K.H.F	Chapman and Hall, London	1999

SEMESTER S6

BIOMASS MANAGEMENT FOR FODDER AND ENERGY

Course Code	OEAGT832	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart to fundamental knowledge on the importance of bio-resources, bio-energy and reactors.
2. To enable the students to gain a sound understanding about residue management in agriculture fields.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to biomass management, biomass resource assessment management techniques, supply chains. Processing of paddy straw, densification- Extrusion process, pellets, mills and cubes, Bailing-classification, uses.	9
2	Residue management for surface mulch and soil incorporation, Paddy Straw choppers and spreaders as an attachment to combine Harvester, Mulch seeder.	9
3	Paddy Straw Chopper-cum-Loader, Baler for collection of straw; Processing of straw/ fodder for animal use.	9

4	Agricultural and horticultural use, cushioning material for fruits and vegetables, Mulching and Composting, Paper and cardboard manufacturing, Straw as a fuel.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the importance of bio resources.	K1
CO2	Gain knowledge and skills on bio energy source technology.	K2
CO3	Understand the importance of biomass in agriculture fields.	K1
CO4	Have a knowledge on paddy straw chopper and balers.	K2
CO5	Gain skill about residue management in agriculture fields.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		2	1			1					2
CO2	3	3	2	1	1		1					1
CO3	3	1	2	1	2		1					2
CO4	3	2	2	2	2		1					2
CO5	3		1				1					2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Food, Feed and Fuel from Bio mass	Chahal, D.S.	IBH Publishing . Pvt. Ltd. NewDelhi	2005
2	Bio Technology and other Alternative Technologies for Utilisation of Bio mass/Agrl.Wastes	Chakravarty, A	Oxford & IBH Pub.Co.Pvt Ltd	2004
3	Bio Chemical Engineering	Alba S. A.E. Humphery and N.E. Milles	John Wiley and sons, New York	2001
4	Bio Chemical Engineering Fundamentals	Baily, J.E and D.F. Ollies	Prescott and Dunn	1999